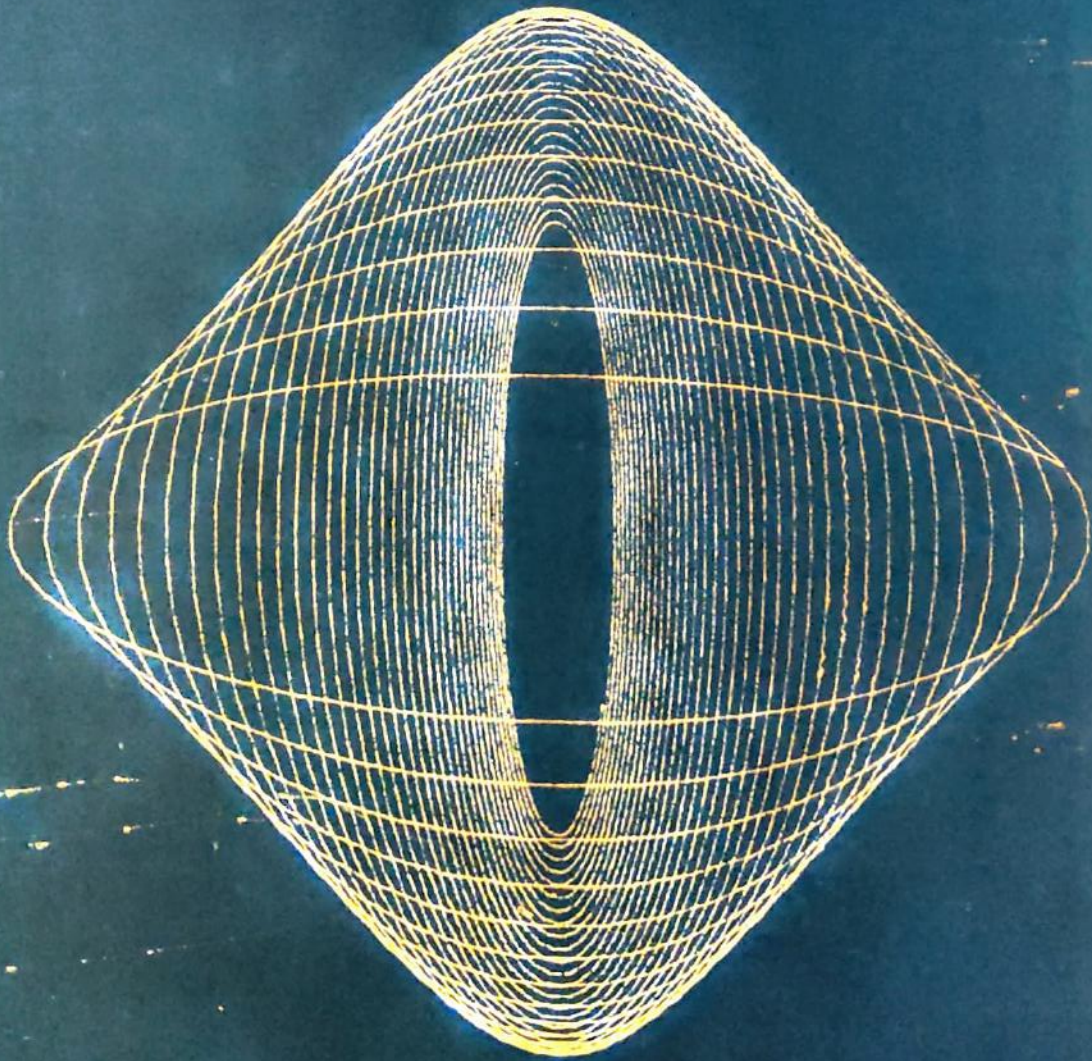


MATH SURVEY



STUYVESANT H.S.

SPRING 1978

We dedicate this issue to Mr. Barry Glotzer for his devotion, not only to our Math Team, but to all the activities of the department, and for his outstanding work as a teacher.

Math Survey

Stuyvesant's Math Magazine

Spring 1978 Volume LIII

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MATRICES & VECTORS

MICHAEL LEV

Little Jimmy spun dizzily, tottering at the edge of his polyfoam couch, as Professor Marty Euclidean eased the ship into hyperdrive. The stars on the video screen went blank. Little Jimmy, being a sixth grader, had only just completed a course in differential equations. As yet he was unfamiliar with the intricacies of inter-dimensional travel. "What in Gauss's name!" exclaimed Jim as he regained his equilibrium.

"We're about to enter a non-euclidean vector sub-space," explained the Professor, very matter-of-factly.

Little Jimmy reached for his zero gravity jacks, but decided that, for the time being, he had better humor the Professor. "That's nice," he said, "but what does it have to do with space — and why did the stars go out?"

"Let me give you an example," replied Euclidean. "Just as from Newton's laws most of classical physics can be derived, from a few basic axioms the properties of every type of space can be extrapolated. Now, a vector space can be very abstract mathematically, and if you're really interested you can read any linear algebra book."

Little Jimmy did not seem really interested.

"Since I see that you'd like to learn more," continued Marty, "I will say that the concepts of linear dependence and spanning set allow us to define a basis, a set of vectors which can generate a vector space V . Now for our example. In the physical world we take our set V to be $\mathbb{R}^n$, the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ of real numbers. Thus $\mathbb{R}^2$ is the set of ordered pairs (a, b) . It's probably obvious to you that this is simply the cartesian coordinate plane you learn about in math 1." Professor Euclidean took out a crumpled gum wrapper and drew:

"In vector space $\mathbb{R}^2$, $(1, 1)$ represents a vector. In the Euclidean cartesian coordinate plane, which obeys the laws of normal geometry, the length of the vector (a, b) is defined as its distance from the origin and can be found by using the Pythagorean Theorem. The basis vectors in $\mathbb{R}^2$ are $A_1 = (1, 0)$ and $A_2 = (0, 1)$, since if r and s are any scalars, from the expression $rA_1 + sA_2$, any other vector in $\mathbb{R}^2$ can of course be generated."

"Of course," repeated Jimmy.

"It can be shown that all possible bases of a vector space have the same number of elements. This number is said to be the dimension of V ."

"I suppose that our space is itself a vector space—but what made the stars go out?"

"Basically true," continued the Professor, delighted with his captive audience. "To be more precise, we live in a metric vector space, also known as an inner product space. A metric vector space is defined as a vector space together with an additional special operation called a SYMMETRIC BILINEAR FORM (S.B.F.). This is a function, satisfying certain conditions, which takes 2 vectors, and assigns to them a real number. As you might remember from AP Physics at Stuyvesant, this bilinear form in the Euclidean plane is

called the dot product, and is defined as: $(a,b) \cdot (c,d) = ac + bd$." Marty punched a button. Diagram 2 appeared on the video screen.

"From the dot product, many properties of the rectangular plane can be derived, including the ideas of perpendicularity and distance. In any metric vector space the length of a vector A is defined as $|A| = \sqrt{A \cdot A}$, where $\cdot$ is any S.B.F. Let us apply this to a vector (a,b) in $\mathbb{R}^2$, using the standard dot product.

$$A = |(a,b)| = \sqrt{(a,b) \cdot (a,b)} = \sqrt{a^2 + b^2}.$$

This is in exact agreement with the FAMILIAR PYTHAGOREAN THEOREM.

In general, two vectors are defined to be perpendicular if

$$A \cdot B = 0.$$

"I guess you're going to give an example of this," interrupted Jimmy.

"Let me give you an example of this," the Professor went on.

"Consider the vectors $(a,b) \cdot (c,d) = 0$. This vector dot product will be precisely in agreement with our normal conception of perpendicularity."

"Why did the stars go out," persisted Little Jimmy, becoming mildly impatient with the Professor. It seemed that he would never get to play with his zero gravity jacks.

"Now we're ready to discuss the stars," explained Professor Euclidean, unphased by his young protégés enthusiasm. "When I put the ship into hyperdrive, it began the complex process of actually changing the S.B.F. of the surrounding space. If I had wished, I could have programmed the computer to change the fabric of space only around us alone, or even around a single object. Here, let me show you. Cut out a small circle from that piece of paper." Marty placed the paper circle next to the glowing control console, and typed $G = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ into the teleprinter.

Suddenly the paper seemed to twist inside out and become hyperbolic in shape!

"I merely changed the bilinear form of the space around the paper from $(a,b) \cdot (c,d) = ac + bd$ to $(a,b) \cdot (c,d) = ac - bd$. This new metric vector space is called the Lorentz Plane (diagram 4). It is easily shown from the definition of perpendicularity that all vectors along the lines $x+y=0$ and $x-y=0$ are $\perp$ to themselves. Furthermore, it is also quite easy to demonstrate why the circle became a hyperbola. I suggest you try to prove this yourself, keeping in mind that a circle is defined as the locus of points equidistant from a fixed origin and that distance $= \sqrt{A \cdot A}$."

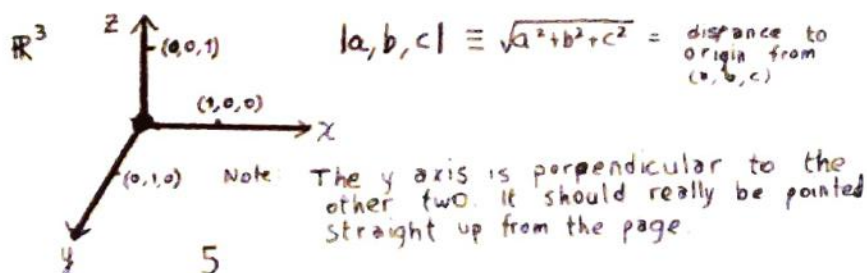
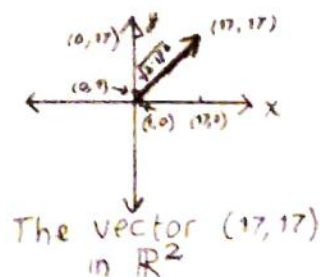
At this point a strange change came over Little Jimmy. He seemed to become genuinely interested.

"Professor, before you said that an S.B.F. must satisfy certain conditions. What exactly are they?"

"The bilinearity property states that: (1) $(A + B) \cdot C = A \cdot C + B \cdot C$ and (2) $(aA) \cdot B = a(A \cdot B)$. The symmetry condition says that: (3) $A \cdot B = B \cdot A$."

"Hmmm. And what about the matrix you typed into the console?"

"Now we come to the interesting part. Consider a symmetric matrix G . As you know, a symmetric matrix is one that is equal to its



own transpose, or $g_{ij} = g_{ji}$ for each element g . Let's also consider an n -dimensional vector space V with basis $A_1, A_2, \dots, A_n$. Just as I showed for $\mathbb{R}^2$ and $\mathbb{R}^3$, any vector A in V can be expressed as $A = a_1 A_1 + \dots + a_n A_n$

as $A = a_1 A_1 + \dots + a_n A_n = \sum_{i=1}^n a_i A_i$. Similarly, we can have another

vector $B = \sum_{j=1}^n b_j A_j$. We can now express a new inner product

vector $B = \sum_{j=1}^n b_j A_j$. We can now express a new inner product $A \cdot B$ as

$A \cdot B = \sum_{i,j=1}^n g_{ij} a_i b_j$, where i and j run from 1 to n in all combinations.

Now let me show you how this applies to Euclidean space. Let

$$G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A = (a_1, a_2, a_3) = (a_1, 0, 0) + (0, a_2, 0) + (0, 0, a_3) \\ = a_1 A_1 + a_2 A_2 + a_3 A_3,$$

and $B = b_1 A_1 + b_2 A_2 + b_3 A_3$. Then,

$$A \cdot B = \sum_{i,j=1}^n a_i b_j g_{ij} = g_{11} a_1 b_1 + g_{12} a_1 b_2 + g_{21} a_2 b_1 + \dots + g_{33} a_3 b_3 =$$

$= a_1 b_1 + a_2 b_2 + a_3 b_3$. This is exactly our definition of dot product for 3-dimensional space!"

"So when you typed in $G = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, the S.B.F. for the vectors (a_1, a_2) and (b_1, b_2) became

$$A \cdot B = \sum_{i,j=1}^n g_{ij} a_i b_j = a_1 b_1 - a_2 b_2."$$

"That's correct. Let me show you some of the other matrices the computer has stored." Marty typed RCL MTRX'S 1-3 into the teleprinter, and the following flashed onto viewscreen 5.

$$G_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

MINKOWSKI
4-SPACE

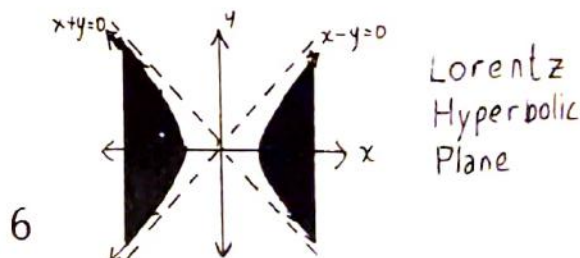
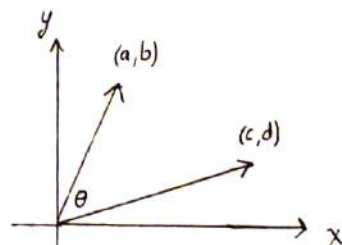
$$G_2 = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -1 \end{bmatrix}$$

NEGATIVE DEFINITE
N-SPACE

$$G_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

MINKOWSKI-LORENTZ
3-SPACE

"Since you've already completed your requirements in relativity, the physical meaning G_1 should be obvious to you," continued the professor. "In the vector space with bases A_1, A_2, A_3 , and A_4 , A_4 can be thought of as a time coordinate. Furthermore, the subspace of G_1 formed by A_1, A_2 , and A_3 is regular Euclidean space. Also, the subspace formed by A_2, A_3 and A_4



is really G_3 ." The professor now typed RCL G_3 : SPHERE. On screen 6 this diagram appeared.

"I'll leave it to you to figure out all the intricacies of this figure. It is fully discussed in the book Metric Affine Geometry, along with the problem of isometries between metric vector spaces. It can easily be shown that every inner product can be expressed as a matrix, and visa-versa. Furthermore, the matrix corresponding to an n-dimensional vector space has dimensions n by n. We are now faced with the problem of classifying all possible vector spaces of dimension n, keeping in mind that some of these are isometric, or equivalent. This problem has been solved as a consequence of Sylvester's Theorem. Sylvester states that, given a matrix G, with zeroes everywhere, except on the main diagonal which has r ones, s negative ones, and t zeroes ($r + s + t = n$), EVERY METRIC VECTOR SPACE OF DIMENSION N IS ISOMETRIC TO ONE OF THESE MATRICES."

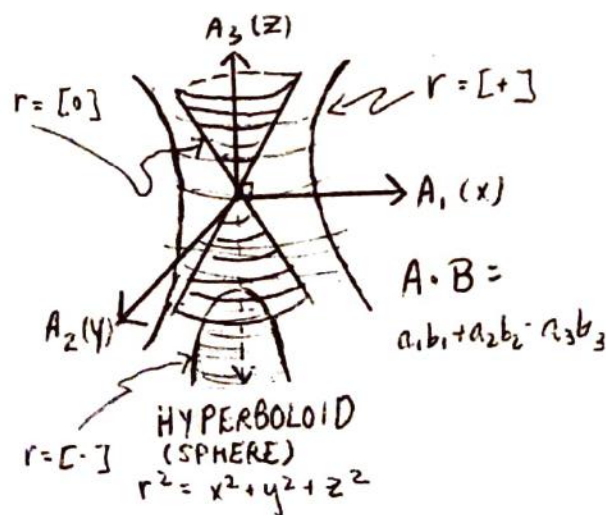
Little Jimmy looked across the room. His small paper circle was still hyperbolic, although the S.B.F. had long ago been shut down. He questioned Professor Euclidean about that.

"It's really very simple. The field, you see, remains about an object for a period of time directly proportional to the object's mass. The Hurwitz-Jacobi conversion formula, which..."

Little Jimmy had an idea. All this was very interesting, but there was only so much he could take in one day. He walked over to the teleprinter and typed PROF. M. EUCLIDEAN:

$G = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Marty stopped his dissertation long enough to shout "NULL SPACE!" before disappearing into a point.

Little Jimmy sat down and played with his zero gravity jacks.



$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 & 0 \\ & 1 & 0 & 0 & \dots & 0 & 0 \\ & & 1 & 0 & \dots & 0 & 0 \\ & & & -1 & \dots & 0 & 0 \\ & & & & -1 & \dots & 0 \\ & & & & & -1 & 0 \\ & & & & & & 0 & 0 & 0 \end{bmatrix}$$

MARKOV CHAINS

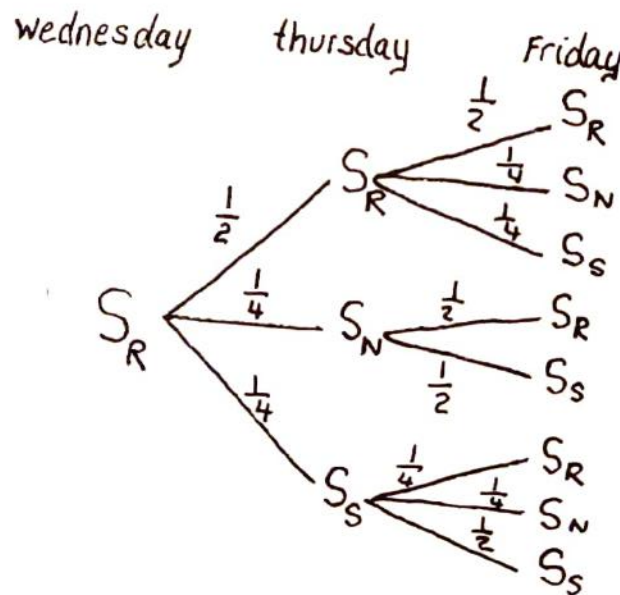
JULIE PAN

During their stay in the Land of Oz, Dorothy and Toto, who were accustomed to Kansas' beautiful climate, noticed that the weather followed a peculiar pattern. The scarecrow, a mathematically inclined strawbrain (yet a kindly fellow), told her, "Dorothy, when the Wiz first arrived, he decreed that there would never be two nice (N) days in a row. If it rained (R) or snowed (S), there was a half chance of it R or S again (correspondingly) the next day, and equal possibilities that it would be N/S or N/R correspondingly." OR:

$$B = \begin{matrix} & \begin{matrix} S(R) & S(N) & S(S) \end{matrix} \\ \begin{matrix} S(R) \\ S(N) \\ S(S) \end{matrix} & \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix} \end{matrix}$$

where i =rows, j =columns
and each P_{ij} =probability
of moving from $S(i)$ to $S(j)$.
This is actually a regular
and ergodic Markov chain.

"Clearly, you can predict the weather quite nicely", lectured the Scarecrow. "I call my diagram transition matrix B." "Well", Dorothy asked, "what will Friday be like?" "Ahh, then let's see the probabilities- Today is Wednesday and it's raining, so:



3/16 for nice, 7/16 for rain, 6/16 for snow."

"Oh this is too much work!" exclaimed Dorothy. "You see, Toto and I want to go on a picnic on Friday." "OK, let's see it this way"

$$\begin{aligned}
2) \quad P(\text{Fri} = S_N) &= \sum_{K=R,N,S} P(\text{Th} = S_K \text{ and Fri} = S_N) \\
&= \sum_{K=R,N,S} P(\text{Th} = S_K) \cdot P(\text{Fri} = N \mid \text{Th} = S_K) \\
&= \sum_{K=R,N,S} P(\text{Th} = S_K) \cdot P_{KN} \\
3) \quad &= P(\text{Th} = S_R) \cdot P_{RN} + P(\text{Th} = S_N) \cdot P_{NN} + \\
&\quad + P(\text{Th} = S_S) P_{SN}
\end{aligned}$$

"You see, P_{KN} 's can be gotten from my transition matrix B. To find $P(\text{Th} = S_K)$ you repeat the process:

$$\begin{aligned}
P(\text{Th} = S_K) &= \sum_{K=R,N,S} P(\text{Wed} = S_K) \cdot P(\text{Th} = S_R) \text{ and since today} \\
\text{is raining,} \quad &= P(\text{Wed} = S_R) \cdot P(\text{Th} = S_R) + 0 + 0 \\
&= 1 \cdot \frac{1}{2} + 0 + 0 = \frac{1}{2}
\end{aligned}$$

Similarly, $P(\text{Th} = S_N) = \frac{1}{4}$

$P(\text{Th} = S_S) = \frac{1}{4}$. Plugging all this into (3),

$$P(\text{Fri} = S_N) = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{4} \cdot 0 + \frac{1}{4} \cdot \frac{1}{4} = 3/16, \text{ which agrees with (1)."}$$

"That is still such a mess!" complained Dorothy. "Isn't it possible to generalize this so you can get all the probabilities for various states at the same time?" "Of course! Did Glenda, the Witch of the North, ever tell you about vectors? Let my transition matrix be

$$B = \begin{matrix} & \begin{matrix} S(1) & S(2) & S(3) & \dots & S(n) \end{matrix} \\ \begin{matrix} S(1) \\ S(2) \\ \vdots \\ S(n) \end{matrix} & \begin{pmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n_1 & n_2 & n_3 & \dots & n_n \end{pmatrix} \end{matrix}$$

To find the probability that it will be $S(m)$ on the $t+1$ event(e), you know by definition from (2) that

$$P(e_{t+1} = S(m)) = \sum_{K=1}^n P(e_t = S(k)) P_{km}.$$

To find the probabilities for all $S(k)$ s on the e_{t+1} event, you can use a probability vector V instead of

a single $P(e_{t+1} = S(m))$: Let $V_{t+1} = (P(e_{t+1} = S(1)), \dots, P(e_{t+1} = S(m)))$.

Then, $V_{n+1} = V_n \cdot B$

So for Friday's weather,

$$V_1 = (\frac{1}{2} \ \frac{1}{4} \ \frac{1}{4}) \text{ for tomorrow's weather.}$$

$$V_2 = (V_1)(B) = (\frac{1}{2} \ \frac{1}{4} \ \frac{1}{4}) \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix} = \left(\frac{7}{16} \ \frac{3}{16} \ \frac{3}{8} \right)$$

Observe further:

$$\begin{aligned} V_1 &= V_1 \\ V_2 &= V_1 B \\ V_3 &= V_2 B = V_1 B^2 \end{aligned}$$

$$\begin{aligned} V_4 &= V_1 B^3 \\ &\vdots \\ V_{n+1} &= V_1 B^n \end{aligned}$$

Further, if you look closely at B^n when $n \rightarrow \infty$, $B^n \rightarrow B'$ where every row vector of B' will be identical (all components > 0). In essence, there will be an equal probability vector for any e_t th event, regardless of the starting state. For example, our weather will eventually approach a probability vector $V = (.40 \ .20 \ .40)$ for any day.

$$\begin{matrix} & S_R & S_N & S_S \\ S_R & (.402 & .199 & .398) \\ S_N & (.398 & .203 & .398) \\ S_S & (.398 & .199 & .402) \end{matrix} = B^4$$

As you see, the rows converge quickly or, if you want to be precise about it, you see that

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -4 & 1 \\ -1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 1 \\ 0 & -4 & 1 \\ -1 & 1 & 1 \end{pmatrix}^{-1}$$

(multiply it out if you want to check me) and if

$$X = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -4 & 1 \\ -1 & 1 & 1 \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} B^2 &= (XD X^{-1})(XD X^{-1}) = XD^2 X^{-1} \\ B^3 &= (XD^2 X^{-1})(XD X^{-1}) = XD^3 X^{-1} \end{aligned}$$

(matrices are assoc. in multiplication)

$\vdots$

$$B^n = (XD^{n-1} X^{-1})(XD X^{-1}) = XD^n X^{-1}$$

Clearly, as $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} \begin{pmatrix} \frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 1 \end{pmatrix}^n = \lim_{n \rightarrow \infty} \begin{pmatrix} (\frac{1}{4})^n & 0^n & 0^n \\ 0 & (-\frac{1}{4})^n & 0^n \\ 0 & 0 & (1)^n \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Therefore,

$$\begin{aligned} B^n &= \begin{pmatrix} 1 & 1 & 1 \\ 0 & -4 & 1 \\ -1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 1/10 & -1/5 & -1/10 \\ 1/5 & 1/5 & -2/5 \end{pmatrix} \\ &= \begin{pmatrix} .4 & .2 & .4 \\ .4 & .2 & .4 \\ .4 & .2 & .4 \end{pmatrix} \end{aligned}$$

"Rats", moaned Dorothy. ".80 chance of $S_R + S_S$ is terrible for picnics."

"Well, that's life for you", uttered the Scarecrow, who always had a maxim for every occasion.

MATH TEAM

Stuyvesant's Math Team consists of students who deprive themselves of sleep and television for the sake of mathematics. Each morning, under the guidance of their coach, Mr. Barry M. Glotzer, members of both the senior and junior teams prepare for competitions and attempt to solve interesting problems.

The senior team members are captains Mark Kleiman and Fred Helenius, Michael Lev, Jinku Lee, and Richard Tello. Junior team members are captains Greg Sorkin and Ashfaq Munshi, Irwin Jungreis, Mark Weiss, and Patrick Davin. Both teams finished first in their respective divisions in the Interscholastic Mathematics League Competition.

On March 14, the annual Mathematical Association of America contest was held. Fred Helenius scored highest in the nation with a perfect score of (150). Stuyvesant's team, comprised of Helenius (150), Kleiman (145), and Sorkin (128) was the highest in the nation with a total **score** of 423. In addition, a total of 12 students placed on the MAA honor roll. Four of them, Fred Helenius, Mark Kleiman, Greg Sorkin, and Gregg Patruno were invited to participate in the Seventh United States Mathematical Olympiad, held on May 2.

Anyone who likes math, or wants to like math, is invited to stop by room 517, at 8:00 each morning. Below, and throughout this magazine, are samples of problems that math team members work on.

PROBLEMS

Let decimal numeral 'AAA' represent the positive integer N .

The sum of the prime factors of N is 47. Find A .

The number of dollars in a check was twice the number of cents.

After paying for a purchase of \$30.69, there were as many cents in the change as dollars in the check and as many dollars in the change as cents in the check. Find the amount of the check.

Given: $y = \frac{3^x + 3^{-x}}{2}$. Express 3^x in terms of y .

Find the sum of the infinite series

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} - \frac{1}{32} + \frac{1}{64} - \frac{1}{128} - \frac{1}{256} + \dots$$

where each positive term is followed by exactly two negative terms, and the absolute value of the n^{th} term is $\frac{1}{2^{n-1}}$.

NEWTON'S SUMS

ASHFAQ MUNSHI

The problem of finding the sum of the p^{th} powers of the roots of an equation frequently occurs in math competitions. The solution of the problem in general is given by Newton's Sums, or Waring's Formula.

We will now proceed to derive this formula and show how easily one can evaluate the sums of the p^{th} powers of the roots of a given n^{th} degree equation, provided that p is less than n . Consider the equation:

$$f(x) = x^n + c_1 x^{n-1} + \dots + c_n. \quad (1)$$

Let the roots of this equation be $x_1, x_2, \dots, x_n$.

$$\text{Define } S_k = \sum_{v=1}^n x_v^k. \quad (2)$$

That is, S_k is the sum of the k^{th} powers of the roots.

$$\text{Rewriting (1), we have } f(x) = (x-x_1)(x-x_2)\dots(x-x_n) \quad (3)$$

Taking the natural logarithm of both sides yields:

$$\ln(f(x)) = \ln(x-x_1)(x-x_2)\dots(x-x_n) = \sum_{i=1}^n \ln(x-x_i).$$

Taking the derivative of (1), we have:

$$f'(x) = nx^{n-1} + (n-1)c_1 x^{n-2} + \dots + c_{n-1}.$$

Alternatively, differentiating (3) yields:

$$\frac{f'(x)}{f(x)} = \sum_{j=1}^n \frac{1}{x-x_j} \quad (4)$$

$$\text{But } \frac{1}{x-x_j} = \frac{1}{x} \frac{1}{1-x_j/x}.$$

Substituting this, (4) becomes :

$$f'(x) = f(x) \sum_{j=1}^n \frac{1}{x} \frac{1}{1-x_j/x}. \quad (5)$$

$$\frac{1}{1-x_j/x} = 1 + \left(\frac{x_j}{x}\right) + \left(\frac{x_j}{x}\right)^2 + \dots + \left(\frac{x_j}{x}\right)^n + \dots = \sum_{v=0}^{\infty} \frac{x_j^v}{x^v}.$$

Therefore, (5) is equivalent to

$$f'(x) = f(x) \sum_{j=1}^n \frac{1}{x} \sum_{v=0}^{\infty} \frac{x_j^v}{x^v}, \quad \text{or}$$

$$f'(x) = \frac{1}{x} f(x) \sum_{j=1}^n \sum_{v=0}^{\infty} \frac{x_j^v}{x^v}$$

$$x f'(x) = f(x) \sum_{j=1}^n \sum_{v=0}^{\infty} \frac{x_j^v}{x^v} = f(x) \sum_{v=0}^{\infty} \sum_{j=1}^n \frac{x_j^v}{x^v},$$

$$\text{but by (2)} \sum_{j=1}^n x_j^v = S_v,$$

$$\text{thus, } x f'(x) = f(x) \sum_{v=0}^{\infty} \frac{S_v}{x^v} \dots\dots (6)$$

From the above, we have:

$$x \cdot f'(x) = nx^n + (n-1)c_1 x^{n-1} + \dots + c_{n-1}x$$

so that the coefficient of x^{n-p} is $(n-p)c_p$.

Expanding the right side of (6), using the fact that

$$f(x) = \sum_{i=0}^n c_i x^i \quad \text{we have}$$

$$x f'(x) = \sum_i \sum_v c_i S_v x^{i-v}.$$

Equating coefficients of x^{n-p} we have

$$(n-p)c_p = \left(\sum_{k=0}^{p-1} c_k S_{p-k} \right) + nc_p \quad (7)$$

Simplifying, we have

$$S_p + c_1 S_{p-1} + c_2 S_{p-2} + \dots + pc_p = 0 \quad (8)$$

Thus we have produced a recursive formula for the sum of the p^{th} powers of the roots of a given equation.

Applications: Using (8), the sum of the first powers of the roots of an equation is

$$S_1 + c_1 = 0, \quad \text{or } S_1 = -c_1.$$

Similarly, the sum of the squares of the roots is

$$S_2 + c_1 S_1 + 2c_2 = 0, \quad \text{or}$$

$$S_2 = c_1^2 - 2c_2, \quad \text{and so on.}$$

We can confirm these results by trying a few numerical examples. These are left for the reader.

PROBLEMS

Express the following sum as a single reduced fraction:

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots + \frac{1}{(2n-1)(2n+1)} + \dots + \frac{1}{49 \cdot 51}$$

Solve for all positive integral values of x, y, z if

$$x + 3y = 11 \quad \text{and} \quad x + y = 3^z.$$

Give answer(s) in the form (x, y, z) .

$\overline{AB}$ is the diameter of a semicircle with center O and with $OA = 30$ inches. Two vertices of a square are on $\overline{AB}$ and the other two are on the semicircle. Find, in square inches, the area of the square region.

A plane region is bounded by the graphs of $x^2 = 4$ and $y^2 = 2y$.

Find the area of the subregion (i.e., the set of points within the given region) whose coordinates satisfy

$$xy^2 \leq x^2 y.$$

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GAME THEORY

MARK KLEIMAN

The theory of games is one of the newest branches of mathematics, with applications in a variety of fields such as economics, politics, defense, and labor relations. The idea of treating real-life issues as mathematical problems began in the 1940's and is due largely to the brilliant mathematician John von Neumann. With the economist Oskar Morgenstern, von Neumann expounded the principles of simple games involving few players.

The difficulty inherent in game theory can be seen by examining the "simplest" type of game, a two-person, zero-sum game. In a zero-sum game the total amount of money that the players have at the beginning is equal to the amount they have at the end. So in a two person game, the amount that A wins is equal to the amount that B loses. Each of the two players plays a strategy of his own choice and the amount that the first player wins or loses is found by consulting a table.

		1	2
A	1	5	3
	2	6	2

If A plays strategy 1 and B plays strategy 2,
B pays A 3\$.

The natural question that arises is "How can this game be resolved so that each minimizes the amount that he has to pay or maximizes the amount he will receive." Clearly both players cannot achieve their maximum gain at the same time. What then is the best strategy for each to play? In the example given, we are able to determine a solution by noting that if A plays strategy 1 and B plays strategy 2, A will receive 3 units from B. If A or B changes strategy, he will fare worse. Therefore they should continue playing these strategies for as long as they play the game.

Unfortunately, it is not always possible to solve games in such a manner. Consider the following:

		1	2
A	1	1	3
	2	4	2

To "solve" this game, suppose that A plays strategy 1 with frequency x and strategy 2 with frequency $1 - x$. Also, assume that B plays strategy 1 with frequency y and strategy 2, $1 - y$. Then the expected return for A is

$$xy + 3x(1-y) + 4(1-x)y + 2(1-x)(1-y) = 2 + x + 2y - 4xy = 5/2 - 4(x - \frac{1}{2})(y - \frac{1}{4}).$$

We can see that A's expectation is at least $5/2$ since he can get $5/2$ by playing with $x = \frac{1}{2}$. A cannot expect to get more than $5/2$ since if B plays with $y = \frac{1}{4}$, he limits A to $5/2$. This game then actually has a solution whereby both players can truthfully assure themselves that they are playing the best strategy.

In the general, two-person game, with pay-off matrix:

A

	1	2	...	n
1	a_{11}	a_{12}	...	a_{1n}
2	a_{21}	a_{22}	...	a_{2n}
3	a_{31}		...	a_{3n}
...				
m	a_{m1}	a_{m2}	...	a_{mn}

Let A play strategy $\|x_1, x_2, \dots, x_m\|$, and B $\|y_1, y_2, \dots, y_n\|$. That is, let A play strategies $1, 2, \dots, m$ with frequencies $x_1, x_2, \dots, x_m$ respectively. Then $x_1 + x_2 + \dots + x_m = 1$, $x_1 \geq 0, x_2 \geq 0, \dots, x_m \geq 0$. The same conditions must clearly hold for the y 's. Then the expected pay-off for A is

$$\sum_{j=1}^n \sum_{i=1}^m a_{ij} x_i y_j = \sum_{j=1}^n y_j \sum_{i=1}^m a_{ij} x_i.$$

The famous minimax theorem of Von Neumann states that if A acts to make his minimum possible winnings as large as possible, that is, if he tries to maximize $\sum_{i=1}^m a_{ij} x_i$ for $j=1, 2, \dots, n$, and if B tries to minimize his greatest possible loss, $\sum_{j=1}^n a_{ij} y_j$ for $i=1, 2, \dots, m$ then both will settle upon a common value. In other words, A can always make sure that he will win at least a certain amount k and B can make sure that he does not lose more than that same amount k . Consequently, all two-person, zero-sum rectangular games are resolvable. For example, consider a game where each player has three choices.

	1	2	3
1	1	-1	-1
2	-1	-1	3
3	-1	2	-1

If A plays $\|6/13, 3/13, 4/13\|$ and B plays $\|6/13, 4/13, 3/13\|$, the expected gain for A is $-1/13$ and B will win $1/13$ on the average.

The minimax theorem proves that any two person, zero-sum game is resolvable, although it may be technically unfeasible to find the proper strategies for A and B even with the use of a high-speed computer.

HORNER'S METHOD

MARK WEISS

How can we solve a cubic equation? The method taught in this school is to try values and hope we come up with an answer. In general, if $ax^3 + bx^2 + cx + d = 0$ has x as a rational root, x must be in the form $\frac{p}{q}$ where p is a factor of d and q is a factor of a . But what if there are no rational roots? One way is to use the Cardano Solution. But, this requires taking square and cube roots of fractional, irrational, and sometimes, complex numbers. So, is there a simpler method? The answer is yes, and it is known as Horner's Method. But before we can use Horner's Method, we must know how to transform an equation to decrease its roots. Consider the equation

$$f(x) = a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = 0 \quad (1)$$

If we substitute $x = (X + H)$ in $f(x) = 0$, we obtain

$f(X+H) = a_0(X+H)^n + a_1(X+H)^{n-1} + \dots + a_{n-1}(X+H) + a_n = 0$ (2)
which is an equation in X whose roots are those of $f(x) = 0$ each reduced by the constant H . If each power of $(X+H)$ in (2) is expanded, the term in X^n will be a_0X^n and on collecting like powers of X , we shall obtain

$$f(X+H) = a_0X^n + A_1X^{n-1} + \dots + A_{n-1}X + A_n = 0 \quad (3)$$

RULE 1 --- To obtain the coefficients $A_1, A_2, \dots, A_n$ in the equation $f(X+H)=0$ whose roots are those of $f(x)=0$ each decreased by H , divide $f(x)$ by $(x-H)$; divide the resulting quotient by $(x-H)$ etc... Continue to divide each resulting quotient by $(x-H)$ until n divisions have been performed. The remainder obtained in the first division is A_n ; the second remainder is A_{n-1} ; the last remainder is A_1 ; the last quotient is a_0 .

Proof of Rule 1 :

From (3),

$$f(X+H) = a_0X^n + A_1X^{n-1} + \dots + A_{n-1}X + A_n \quad (4)$$

In (4), substitute $X=x-H$; then $f(X+H)$ becomes $f(x)$, and we obtain

$$f(x) = a_0(x-H)^n + A_1(x-H)^{n-1} + \dots + A_{n-1}(x-H) + A_n \quad (5)$$

From (5), if we divide $f(x)$ by $(x-H)$ the quotient is

$$a_0(x-H)^{n-1} + A_1(x-H)^{n-2} + \dots + A_{n-2}(x-H) + A_{n-1} \quad (6)$$

On dividing (6) by $(x-H)$ the new quotient obtained is

$$a_0(x-H)^{n-2} + A_1(x-H)^{n-3} + \dots + A_{n-3}(x-H) + A_{n-2}$$

and the remainder is A_{n-1} continuing through n divisions as specified in Rule 1, the last quotient is a_0 and the remainders in the order obtained, are $A_n, A_{n-1}, \dots, A_2, A_1$, which is what we wanted.

Horner's Method for finding a positive irrational root follows:

(a) Locate the root between successive integer. The smaller integer is the integral part of the root.

(b) Transform $f(x)$ into an equation $f_1(x_1)$ whose roots are those of $f(x)$ decreased by the smaller of the integers of Step 1. Then $f_1(x_1)$ has a root between 0 and 1. Locate this root between successive

tenths. The smaller tenth is the tenths part of the root.

(c) Transform $f_1(x_1)$ into an equation $f_2(x_2) = 0$ whose roots are those of $f_1(x_1) = 0$ decreased by the smaller of successive tenths found in Step 2. Then $f_2(x_2)$ has a root between 0 and .1.

Locate this root between successive hundredths. The smaller hundredth is the hundredth part of the root.

(d) Continue this process, to obtain any desired accuracy. Each step yields, accurately, one more decimal place of the root.

MATH SOCIETY

The Math Society, in its program to provide the students of Stuyvesant High School with exposure to mathematical topics not normally found in the school curriculum, has met with great success this year. Its lectures have been both informative and varied, and as in the past we have been fortunate in attracting many top speakers. The topics discussed have ranged from

Crazy Sets in the Unit Interval to Graph Theory, and have been presented by such distinguished professors as Professors Lax and Novikoff of N.Y.U.'s Courant Institute and S. Winograd of I.B.M.'s Thomas J. Watson Research Center. Society members have also enjoyed talks by Professors Wernick and Zirkel of CUNY and Nassau Community College, respectively, and Professor David C. Kelly of Hampshire College who told us "How Far You Can See in an Apple Orchard."

As well as hearing guest lecturers speak, our many club participants found the film "Donald in Mathmagic Land" to be both entertaining and informative. In addition, students were introduced to opportunities for further study in mathematics at a special meeting which discussed summer science programs.

Math Society president Michael Lev and vice-presidents Ashfaq Munshi and Steven Lev, with the aid of faculty advisor Stuart Weinberg have organized this year's club activities. Also, Greg Sorkin has been especially helpful in obtaining speakers. The Math Society's series of meetings this year have met with great approval and we feel that many students have benefited from them. We hope that they continue to do so and encourage all interested students to attend.

TRANSFINITE NUMBERS

PIERRE BORCZUK

Georg Cantor (1845-1918) was a mathematician who founded the set theory and introduced the concept of the cardinality of infinite sets. These numbers, representing the amount of members in infinite sets, are called transfinite numbers.

The first of the transfinite numbers represents the cardinality of the counting numbers. This is the simplest and "smallest" number. Other sets equivalent to the set of counting numbers are the set of even numbers, the set of odd numbers, the set of integers, and the set of rational numbers. The transfinite number assigned is $\underline{a}$ or $\aleph_0$ (aleph-null). (For proof of the equivalence stated above, see Infinity by Lieber.)

Another transfinite number represents the cardinality of the real numbers. To prove that there are more real numbers than counting numbers:

- 1) We first consider all the real numbers r in the interval $0 < r < 1$
- 2) All these numbers can be expressed as infinite decimal fractions. For example: $1/3 = .333\dots$

- 3) Then, if these infinite decimals are arranged in an ordered sequence, they will be equivalent to the set of natural numbers.

```

1-----0.a11a12a13a14...
2-----0.a21a22a23a24...
3-----0.a31a32a33a34...
4-----0.a41a42a43a44...
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Each a_{ij} (i =row; j =column) is a digit of an infinite decimal fraction.

- 4) But, the above array is not ordered, for we can find one infinite decimal that is not included.

- 5) We construct this decimal $r_1 = .x_1x_2x_3x_4\dots$, x being any digit from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

- 6) Now, a condition is imposed: each x_i cannot be equal to each a_{ii} . This means that the first digit of r_1 cannot be the same as the first digit of $0.a_{11}a_{12}a_{13}\dots$ ($x_1 \neq a_{11}$), so on.

When this is done, the real number r differs from the first infinite decimal in the first place, the second decimal in the second place and---differs from the n^{th} infinite decimal in the n^{th} place. It is not included in the above array. But the array had the cardinality $\underline{a}$. So we assign $\underline{c}$ or $\aleph_1$ to the set of real numbers ($\underline{c} > \underline{a}$).

Let us now use the transfinite cardinal numbers $\underline{a}$ and $\underline{c}$ in the operations of addition, subtraction, multiplication, division, and raising to a power.

I Addition-

- a) $\underline{a} + \underline{a} = \underline{a}$ - the cardinality of the set of even numbers + the cardinality of the set of odd numbers = the cardinality of the set of counting numbers.

b) $\underline{c} + \underline{c} = \underline{c}$ - point set $\{0 < x < 1\}$ + point set $\{1 \leq x < 2\} =$ point set $\{0 < x < 2\}$ which has the cardinality $\underline{c}$.

II. Subtraction- It is not defined using transfinite numbers because different results can be obtained using the same sets. For example:

If $N = \{1, 2, 3, \dots\}$, then $N - N = \{1, 2, 3, \dots\} - \{1, 2, 3, \dots\} = 0$

But if $N_1 = \{2, 4, 6, \dots\}$, then $N - N_1 = \{\overset{\underline{a}}{1}, 2, 3\} - \{\overset{\underline{a}}{2}, 4, 6, \dots\} = \overset{\underline{a}}{\{1, 3, 5, \dots\}}$

III. Multiplication-if $\underline{m}$ is the cardinality of a finite set:

a) $\underline{m} \times \underline{a} = \underline{a}$ because $a + a + a + \dots + a = a$

b) $\underline{m} \times \underline{c} = \underline{c}$ because $c + c + c + \dots + c = c$

c) $\underline{c} \times \underline{c} = \underline{c}$ consider a square of unit length, with one side being the point set $0 < x < 1$ and the adjacent side $0 < y < 1$. Each of these point sets has the cardinality $\underline{c}$. Each x and y can be expressed in the following manner,

$$x = .a_1 a_2 a_3 a_4 \dots \quad y = .b_1 b_2 b_3 b_4 \dots$$

then the coordinate (x, y) is expressed as the infinite decimal z

$$z = .a_1 b_1 a_2 b_2 a_3 b_3 a_4 \dots$$

Since all the points interior to the square form the set of infinite decimals z , $0 < z < 1$, and this set is the same as the set of real numbers, they both have the cardinality $\underline{c}$.

d) $\underline{a} \times \underline{c} = \underline{c}$ because $\underline{m} \leq \underline{a} \leq \underline{c}$

$$\underline{m} \times \underline{c} \leq \underline{a} \times \underline{c} \leq \underline{c} \times \underline{c}$$

$$\underline{c} \leq \underline{a} \times \underline{c} \leq \underline{c} \quad \text{so } \underline{a} \times \underline{c} = \underline{c}$$

IV. Division like subtraction leads to different results using the same sets.

V. Powers of cardinal numbers

$10^a = \underline{c}$ If we have a real number $0.x$, where x_n can be replaced by the digits $0, 1, 2, 3, 4, 5, 6, 7, 8, 9$, there are 10 possible values. In the real $\# 0.x_1 x_2$, there are 10^2 possibilities. Taking $\underline{a}$ decimal places, then we have 10^a different decimals. But the set of all these infinite decimal fractions has the cardinality $\underline{c}$.

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ACHIEVEMENT TEST TIPS

It's that time of year again.....Achievement Tests, SAT's, and an awful lot of anxiety. That's why MATH SURVEY has compiled this list of formulas- so you can breeze through all the math tests with super scores!

Review:

Level I

$$\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$$

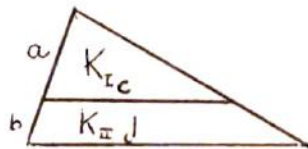
$$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin \theta = \frac{y}{x^2+y^2}, \quad \cos \theta = \frac{x}{x^2+y^2},$$

$$\tan \theta = \frac{x}{y}$$

Minimum/maximum value of $ax^2 + bx + c = 0$ occurs at $x = \frac{-b}{2a}$.

$$\frac{a}{a+b} = \frac{c}{d} \quad \frac{K_I}{K_I+K_{II}} = \frac{c}{d}^2.$$



Sum of a geometric series $a, ar, ar^2, \dots, ar^n$ is $\frac{a(1-r^{n+1})}{1-r}$.

Sum of an infinite geometric series is $\frac{a}{1-r}$.

Sum of an arithmetic progression $a, a+d, \dots, a+(n-1)d$ is :

$$\frac{n}{2}(a+l) \text{ where } l \text{ is the last term, or}$$

$$\frac{n}{2}(2a + (n-1)d).$$

Solution of $ax^2 + bx + c = 0$ is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin (2\theta) = 2\sin \theta \cdot \cos \theta$$

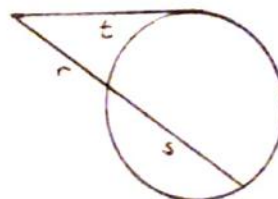
$$\cos (2\theta) = \cos^2 \theta - \sin^2 \theta = 2\cos^2 \theta - 1$$

$$\sin\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$t^2 = r(r + s)$$



LEVEL II

Probability $(A \cup B) = P(A) + P(B) - P(A \cap B)$
 number of ways of choosing K items from N items is:

$$\binom{N}{K} = \frac{N!}{K!(N-K)!}$$

where

$$N! = N(N-1) \dots 1$$

e.g. $5! = (5)(4)(3)(2)(1) = 120$

Number of permutations of N items: $N!$

The product of $a^{(1/2)} a^{(1/4)} a^{(1/8)} \dots a^{(1/2^n)} \dots$

$$= a^{(1/2)+(1/4)+(1/8)+\dots+(1/2^n)+\dots} = a$$

Area of a triangle = $(1/2)ab \cdot \sin C$ or $K = (s(s-a)(s-b)(s-c))^{(1/2)}$
 where $s = (a+b+c)/2$.

Volume of rectangular solid = $(x)(y)(z)$

Volume of sphere = $(4/3)\pi r^3$

Surface area of sphere = $4\pi r^2$

Surface area of rectangular solid = $2(xy + yz + xz)$

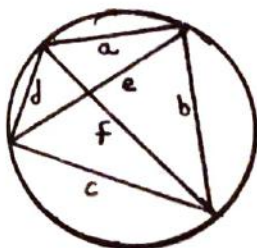
Length of diagonal of rectangular solid = $(x^2 + y^2 + z^2)^{(1/2)}$

Law of sines: $(a/\sin A) = (b/\sin B) = (c/\sin C) = 2R$

Law of cosines: $c^2 = a^2 + b^2 - 2ab \cos C$

If $y = \sin x$ then $x = \arcsin y$.

Similarly $y = \cos x$ implies $x = \arccos y$; $y = \tan x$ implies $x = \arctan y$



$$ab + cd = ef$$

$\tan(A) + \tan(B) + \tan(C) = \tan(A) \tan(B) \tan(C)$ if $A+B+C = 180^\circ$.

Arithmetic Mean: $\frac{a+b}{2}$

Geometric Mean: $\sqrt{ab}$

Harmonic Mean: $\frac{2ab}{a+b}$

ARITHMETIC mean $\geq$ GEOMETRIC mean $\geq$ HARMONIC mean (A.M.)(H.M.)=(G.M.)

Sum of interior angles of an N sided polygon: $180(N-2)^\circ$

Area of a trapezoid: $\frac{1}{2}h(a+b)$ where a,b are bases, and h is the altitude.

GOOD LUCK!

PASCAL'S THEOREM

RALPH SIEGEL

Blaise Pascal was a French physicist and mathematician who lived in the early seventeenth century. At the age of sixteen, he discovered one of the most important theorem of projective geometry, which today bears his name. The theorem states: A hexagon which is inscribed in a conic section has intersections of its three opposite sides lying on the same straight line. It is the goal of this article to prove this theorem using projective geometry.

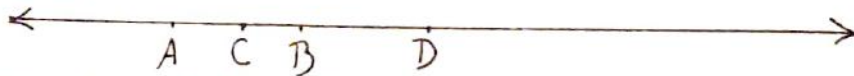
Here are some basic definitions and theorems that will be used in the proof.

pencil- a group of concurrent lines.

range- a group of collinear points.

cross ratio- a specific ratio between the segments set off by four points of a range.

For example:



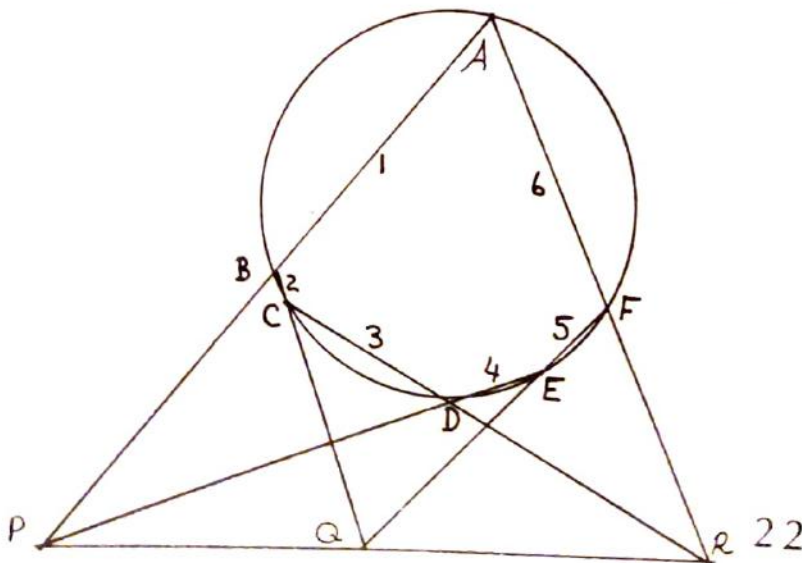
One cross ratio of this line is $(CA/CB)/(DA/DB)$

A projected image-The locus of points formed when, from a given point in space, lines are drawn through a specified primary (planar) image to a given secondary plane.

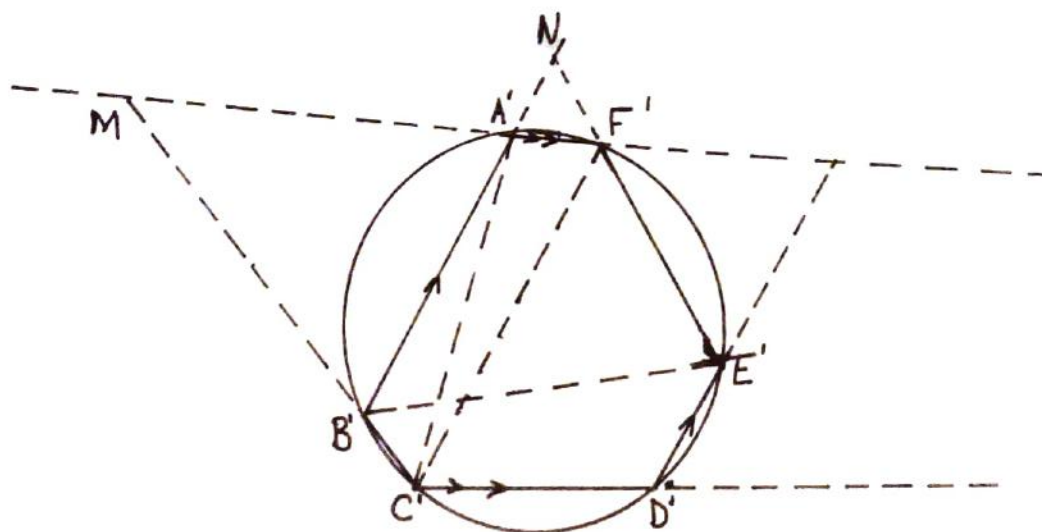
Preliminary theorem-The cross ratio of a pencil formed by lines joining four fixed points and one variable point on a conic is independent of the position of the variable point.

Projected image: The locus of points formed when, from a given point in space, lines are drawn through a specified primary (planar) image to a given secondary plane.

We now begin with figure 1 in our attempt to prove the original theorem. Because of the projective properties of conics, it is enough to prove the theorem only for the case of a circle, since a circle may be projected into a parabola, hyperbola, or an ellipse. The theorem also holds for the degenerative conics.



AB and ED intersect at P, DC and FA at R, and BC and FE at Q. We must prove that PQR is a straight line. To do this we draw figure 2.



We can construct figure 2 by projecting figure 1 from a point X (not in the plane of the circle) onto a plane parallel to the lines PX, and RX. Thus, NB' and D'E' are parallel, as are line segments C'D' and MF'. (Note: Line segments B'C' and A'F' are extended to form point M and line segments B'A' and F'E' are extended to form point N.) We now have to prove that line segments MC' and NE' are parallel.

The pencils C' (B'A'F'D') and E' (B'A'F'D') have the same cross ratio. The former pencil is intercepted by the transversal MF', and has the range (MA'F'∞). This range has the cross ratio F'M / F'A'. The second pencil is intercepted by the transversal NB', and has the range (B'A'N∞). The range has the cross ratio NB' / NA'. Now, if

$$\frac{F'M}{F'A'} = \frac{NB'}{NA'} \quad \text{OR} \quad \frac{F'M - F'A'}{F'A'} = \frac{NB' - NA'}{NA'} = \frac{A'M}{F'A'} = \frac{A'B'}{N A'}$$

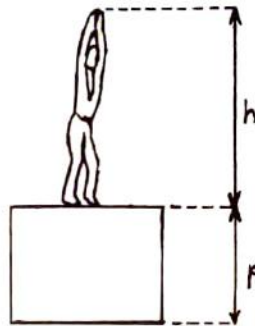
then triangles NA'F' and MA'B' are similar. Therefore NE' and B'C' are parallel. Now since the images of P, Q, and R are collinear at infinity, (figure 2) they must be collinear in figure 1. This completes the proof of the theorem. For a proof of Pascal's dual theorem (by Branchion) and for more information on projective geometry see Introduction to Geometrical Transformations, by Edward H. Barry.

REGIOMONTANUS' PROBLEM

ERIC SINGMAN

A problem similar to the one that will be presented here was first posed in 1471 by the mathematician Johannes Müller, known as Regiomontanus.

PROBLEM:



Suppose that a statue of height h stands on a vertical pedestal of height p . A man whose eye level above the ground is d walks toward the statue gazing at it. At what distance from the base of the statue should he stand in order to make the statue appear the largest (i.e. so that his lines of sight to the bottom and top of the statue subtend the largest angle.)?

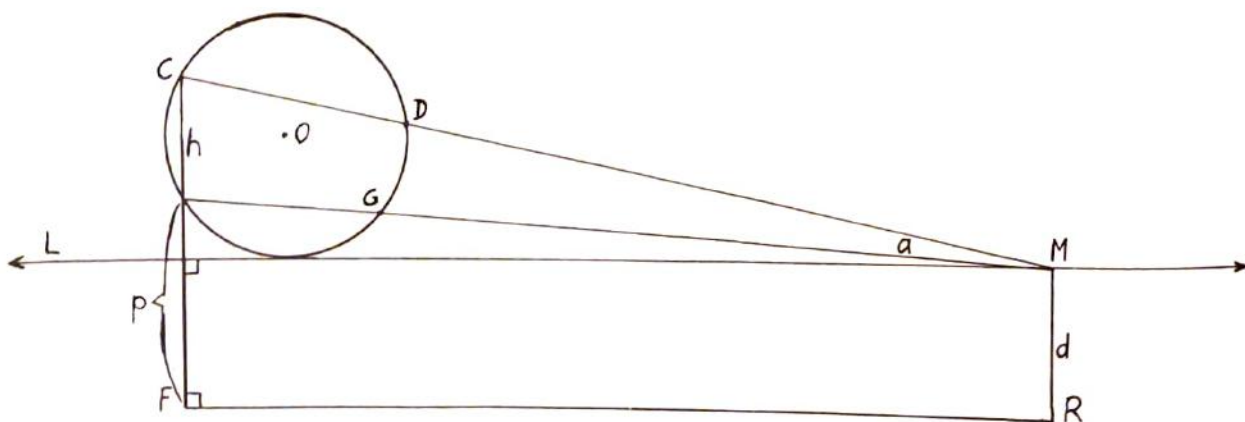
SOLUTION:

We will consider 3 cases:

I) $d < p$

II) $d > p+h$

III) $p < d < p+h$



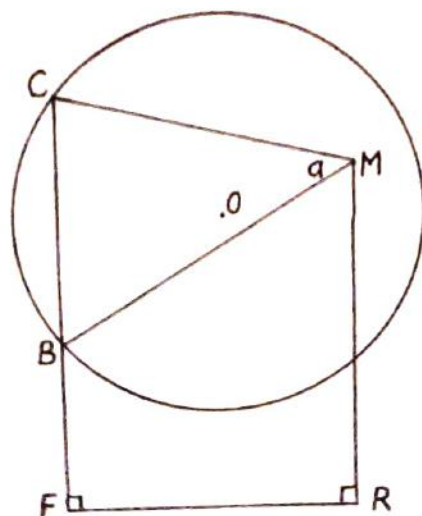
I) In the above diagram, $\overline{CB}$ is the statue, $\overline{BF}$ is the pedestal, L is the locus of the man's eye as he moves along the ground ($\overline{RF}$) and angle BMC is his viewing angle at position R , which has measure a . We therefore wish to maximize a . To do so, let us first construct circle O through points B and C tangent to line L . D is the point where $\overline{CM}$ intersects the circle and G is the point where $\overline{BM}$ intersects the circle. 24

The measure of $\underline{a}$ can be found by subtracting the measure of arc DG from the measure of arc CB and dividing by two; a formula from Euclidean geometry. The measure of arc CB must remain constant because the statue's height, a chord in circle O, will always remain the same. The measure of arc DG can be greater than or equal to zero as the man moves along L. It can't be negative because we are dealing with real lengths. To make $\underline{a}$ as large as possible, we have to make the measure of arc DG as small as possible, namely zero. If this is the case, the value of $\underline{a}$ will be one half the measure of arc CB. This happens to be the measure of an inscribed angle in circle O which subtends arc CB. Therefore, to have the largest viewing angle, the man must stand at the point where the line L is tangent to the circle, because at that point, angle BMC is an inscribed angle. Therefore, the answer to the problem is that he must be as far from the pedestal as the point of tangency of L and circle O.

II) For case #2 , we employ the same method. The proof is left for the reader.

III) For the third case, we begin with a diagram similar to the one used for case #1 , by constructing any circle O through points B and C as shown

It is apparent that the man's viewing angle increases as his distance decreases. Therefore , the maximum value of $\underline{a}$ occurs when he is as near the statue as is possible.



TRISECTION OF AN ANGLE

STEVEN LEV

The three famous geometric construction problems of antiquity are squaring the circle, doubling the cube, and trisecting the angle. If even one angle is found that cannot be trisected, then we can conclude that the statement "An arbitrary angle can be trisected by classical methods (using the unmarked straightedge and a compass only)" is incorrect. We will use as an example the 60° angle. If trisecting a 60° angle is possible, then we could construct a 20° angle. However, if we can construct a 20° angle, it should be possible to construct a regular nonagon, since the central angle of a nonagon is given by

$$\frac{(n-2)(180^\circ)}{n} = \frac{(9-2)(180^\circ)}{9} = 40^\circ = 2(20^\circ).$$

Consider a unit circle in the complex plane, containing the point $(1,0)$. This point can be expressed in polar form as $1(\cos 0^\circ + i \sin 0^\circ)$. By DeMoivre's theorem, the nine ninth roots of one are given by

(A) $\cos \frac{360K^\circ}{9} + i \sin \frac{360K^\circ}{9} = \cos 40K^\circ + i \sin 40K^\circ$,
where $K = 0, 1, 2, \dots, 8$. Each successive value of K corresponds to the points $1, P, P^2, P^3, \dots, P^8$ respectively, forming the vertices of a regular nonagon on the unit circle. (See fig. 1)

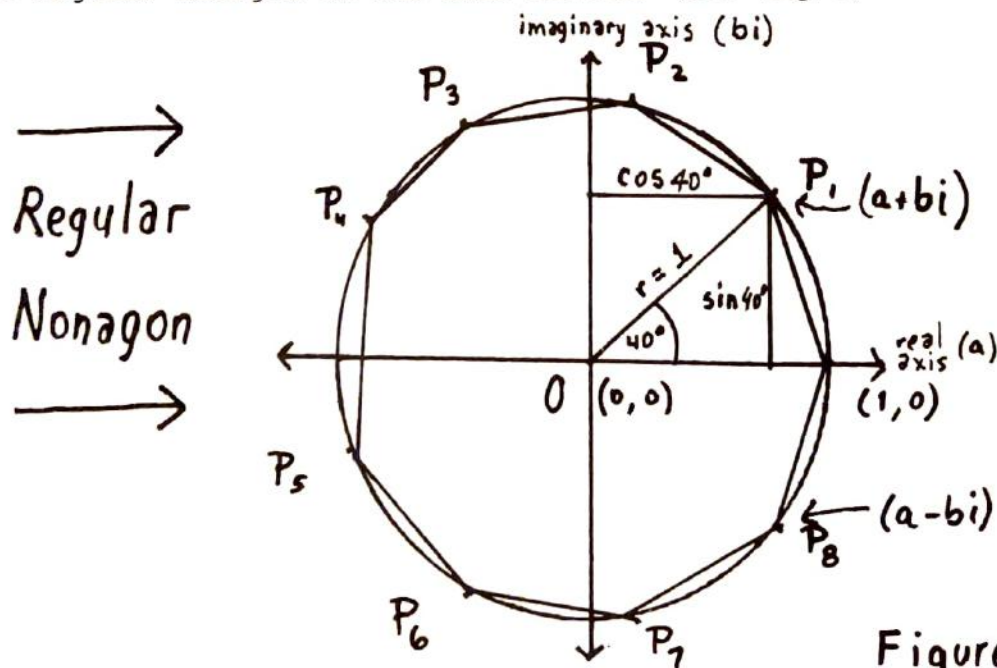
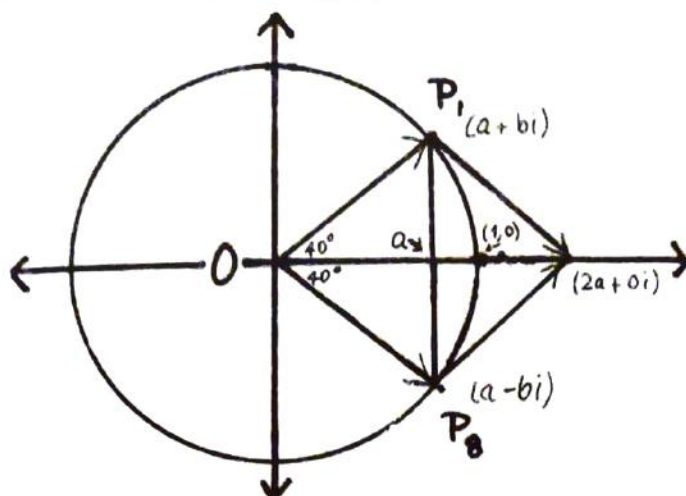


Figure 1

Before beginning, we note that, by definition, $P^9 = 1$. In order to construct the nonagon, we must be able to locate point P . The other vertices can then be found by simply duplicating the chord $\overline{AP}$ around the circumference of the circle. Finding P is equivalent to locating a , the x-coordinate of P , and constructing the perpendicular to $\overline{OA}$ through a . The point where the perpendicular intersects the circle is the point P .

We observe that (see fig.2)



$$P_1 + P_8 = 2a$$

$$a+bi + a-bi = 2a$$

Figure 2

1) $p + p^8 = (a + bi) + (a - bi) = 2a$.

Squaring both sides gives us

2) $p^2 + 2p^9 + p^{16} = 4a^2$.

We know that $p^9 = 1$, so that $2p^9 = 2$ and $p^{16} = p^9 \cdot p^7 = p^7$.

Rewriting equation (2) yields:

3) $p^2 + 2 + p^7 = 4a^2$ or $p^2 + p^7 = 4a^2 - 2$.

If (1) and (3) are multiplied we get

4) $(p + p^8)(p^2 + p^7) = 2a(4a^2 - 2)$, which is the same as $p^3 + p^8 + p^{10} + p^{15} = 8a^3 - 4a$. But, $p^{10} = p^9 p = p$ and $p^{15} = p^9 p^6 = p^6$.

Thus, we have $p^3 + p^8 + p + p^6 = 8a^3 - 4a$. Combining this with equation (1) we find that

4) $p^3 + p^6 = 8a^3 - 6a$

$$p_3 + p_6 = -1$$

$$-\frac{1}{2} + \frac{\sqrt{3}}{2}i + -\frac{1}{2} - \frac{\sqrt{3}}{2}i = -1$$

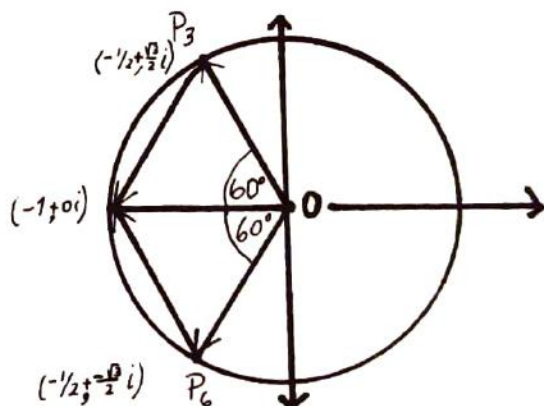


Figure 3

Studying the figure (3) - we see that $p^3 + p^6 = -1$ (addition of complex numbers - let $k=3$ in equation (A)). Therefore, equation (4) becomes $-1 = 8a^3 - 6a$, or, $(8a^3 - 6a + 1 = 0)$.

Solving this using Cardan's formula for the solution of a cubic equation (see Math Survey 1975), we find that a is only expressible in terms of cube roots. However, it is only possible to perform five operations (square roots, addition, subtraction, multiplication, and division) on given line segments using the classical tools of geometry. Therefore, it is impossible to find a , as we would have to construct the cube root of a line segment. Thus a regular nonagon cannot be formed, and it follows that a 60° angle cannot be trisected.

STEINER'S PROOF

FRED HELENIUS

Problem: What is the largest area which can be enclosed with 100 yards of barbed wire?

Solution: This problem is a special case of the so-called isoperimetric problem: What is the largest plane region having a given perimeter? Clearly, such a region must exist since a given perimeter can only enclose a limited area. The following is based on a proof by Steiner.

To begin the solution of the isoperimetric problem, the notion of convexity of plane regions is defined as follows:

A plane region R is called convex if, for any points A and B contained in R , all the points on line segment AB are contained in R . If, for some points A and B contained in R , some point C on line segment AB is not contained in R , R is called non-convex.

Assume that the solution to the isoperimetric problem is a non-convex region (see diagram 1). Then, by the definition, there must be two points A and B for which some points on AB are not contained in the region. If the area(s) between the region and the portion(s) of AB outside the region is (are) reflected about AB , the perimeter of the newly formed region is the same as that of the original, but its area is greater. (In the diagram, the area in the shaded region has been added). However, this contradicts the given fact that the original region is the solution to the isoperimetric problem. Therefore, the assumption that the region is non-convex is false, and the solution to the isoperimetric problem is necessarily a convex region.

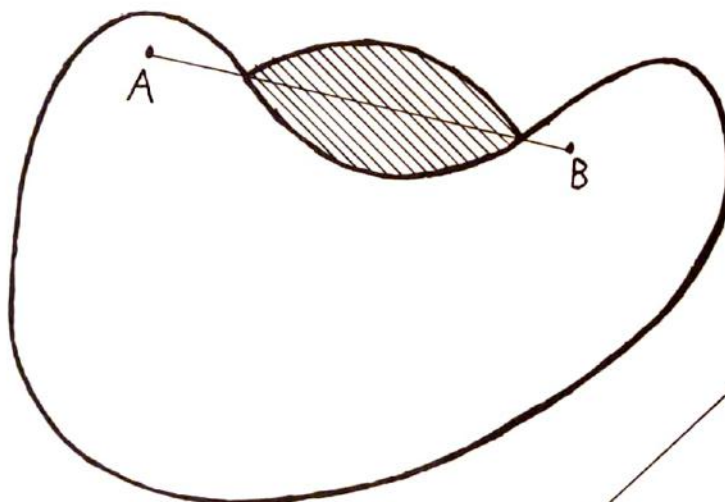


Diagram 1

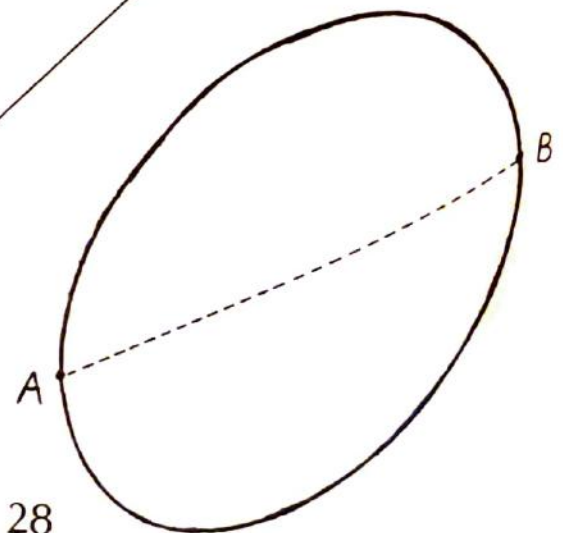


Diagram 2

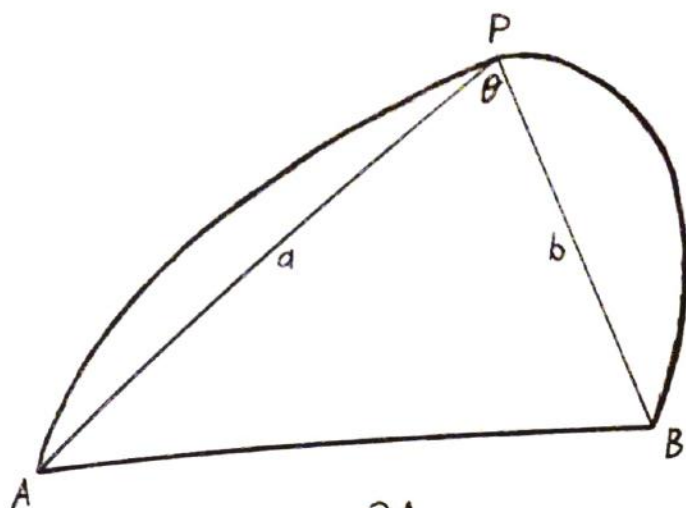


Diagram 3A

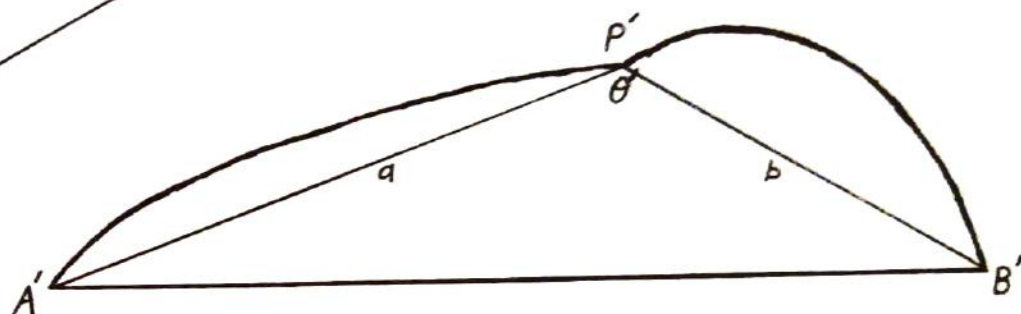


Diagram 3B

If we have the region which solves the problem, a point A on the boundary may be chosen and the point B which is half of the way around the perimeter from A may be located (see dia. 2). Since this region is the solution to the original problem, each half-region must be the largest that can be enclosed by a line segment and an arc of given length (equal to half the perimeter of the entire figure). Suppose that the region in diagram 3A has been proposed as a solution to this secondary problem. Choose a point P in the arc AB, and label $\angle APB = \theta$, $AP = a$, and $BP = b$. A similar figure, with the same length curve enclosing the area, maybe formed by changing the measure of $\angle APB$ to θ' , as in diagram 3B, where the region bounded by $A'P'$ and arc $A'P'$ is congruent to the region bounded by AP and arc AP , and similarly the region bounded by $B'P'$ and arc $B'P'$ is congruent to the region bounded by BP and arc BP . Clearly, the largest figure of this type is formed when the area of $\triangle ABP$ is largest. However, the area of triangle ABP is given by $\frac{1}{2}ab\sin\theta$. Since a and b are constant, the area is maximized when $\theta = 90^\circ$. Hence, the largest region enclosed by an arc of given length and a line segment has all of its inscribed angles equal to right angles. The only arc which fulfills this condition is the semicircle.

Since each half-region is a semicircle, the solution to the isoperimetric problem is a circle with the given perimeter. In the specific problem given, the area enclosed equals $\frac{2500}{\pi}$ square yards.

HARMONOGRAMS

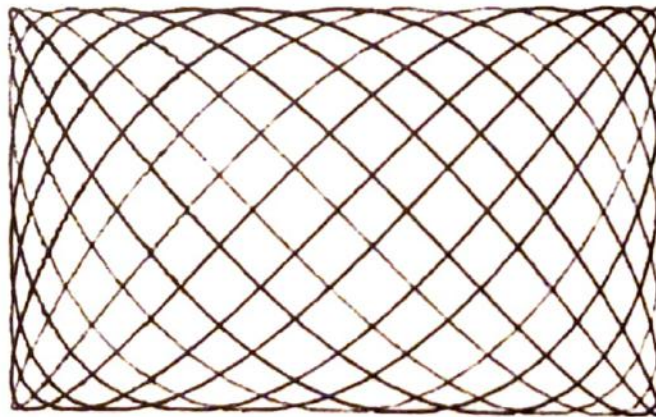
GREG SORKIN

Lissajous figures are a family of curves generated by a pair of sine waves acting at right angles to each other, i.e.

$$x = a_1 \sin(f_1 t + \phi_1),$$

$$y = a_2 \sin(f_2 t + \phi_2),$$

where t is the parameter, f_1 and f_2 are the frequencies of the sine waves, ϕ_1 and ϕ_2 are their initial parameters, and a_1 and a_2 are their amplitudes. This can be easily visualized as a pen² being guided back and forth parallel to the x -axis by one pendulum, and up and down parallel to the y -axis by a second pendulum. The two pendulums may have different amplitudes and frequencies, and may be out of phase with each other. The patterns created are exemplified by the following:



Harmonograms are both more interesting and more complex mathematically than Lissajous figures. They are less strictly defined, and include many interactions of two pendulums. In fact, the simplest class of harmonograms is actually that of the Lissajous figures. The most complex are known as double-elliptic pendulum harmonograms. The apparatus for their construction consists of one pendulum pivoting on the bob of another pendulum. The xy coordinates of either the higher bob (the one which acts as a pivot) or the lower bob (which swings freely) may be traced.

$$\text{By physics, } F_1 = - \frac{m_1 g + m_2 g}{l_1} x_1 + \frac{m_2 g}{l_2} x_2,$$

$$\text{and } F_2 = - \frac{w_2}{l_2} x_2.$$

Since $F = ma$, and $a = x''$,

$$m_1 x_1'' = F_1 = - \frac{m_1 g + m_2 g}{l_1} x_1 + \frac{w_2}{l_2} x_2 ,$$

$$m_2 x_2'' = F_2 = - \frac{w_2}{l_2} x_2 ,$$

$$\text{or } x_1'' = - \frac{m_1 g + m_2 g}{m_1 l_1} x_1 + \frac{m_2 g}{m_1 l_2} x_2 ,$$

$$x_2'' = \frac{-m_2 g}{m_2 l_2} x_2 = - \frac{g}{l_2} x_2$$

$$\text{Letting } K = \begin{pmatrix} -\frac{m_1 g + m_2 g}{m_1 l_1} & \frac{m_2 g}{m_1 l_2} \\ 0 & -\frac{g}{l_2} \end{pmatrix} \quad \text{and} \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

we have $x'' = Kx$.

If $v = x'$ and $w = \begin{pmatrix} x \\ v \end{pmatrix}$, $v' = Kx$, $w' = \begin{pmatrix} x' \\ v' \end{pmatrix} = \begin{pmatrix} v \\ Kx \end{pmatrix} = \begin{pmatrix} 0 & I \\ K & 0 \end{pmatrix} w$,

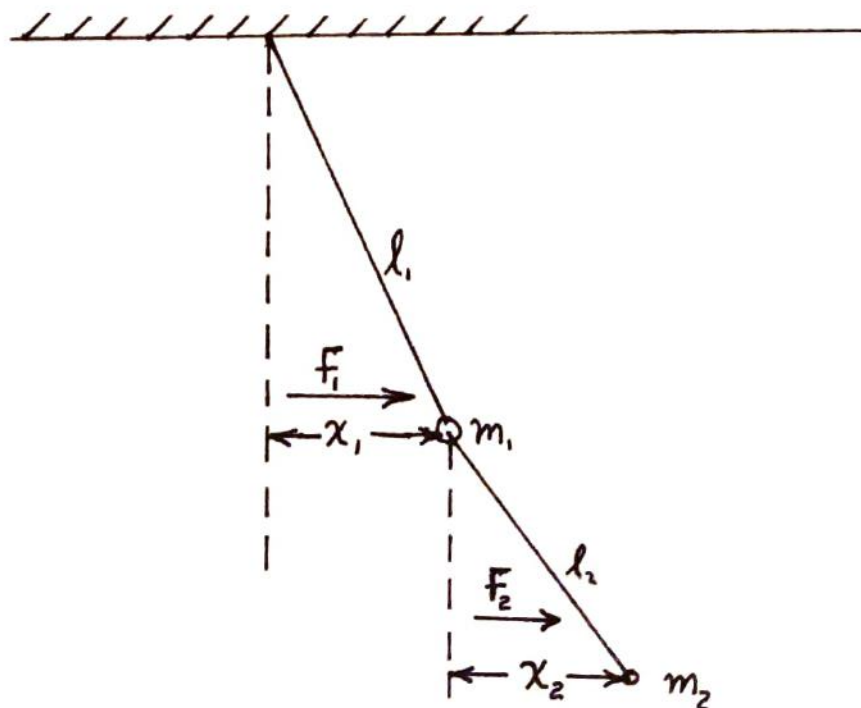
Hence $w = e^{\begin{pmatrix} 0 & I \\ K & 0 \end{pmatrix} t} w(0)$. (Where I is the identity matrix).

Since this really only gives displacements and velocities along the x-axis, we may rewrite this as:

$$\begin{pmatrix} 0 & I \\ K & 0 \end{pmatrix} t \quad \begin{pmatrix} 0 & I \\ K & 0 \end{pmatrix} t$$

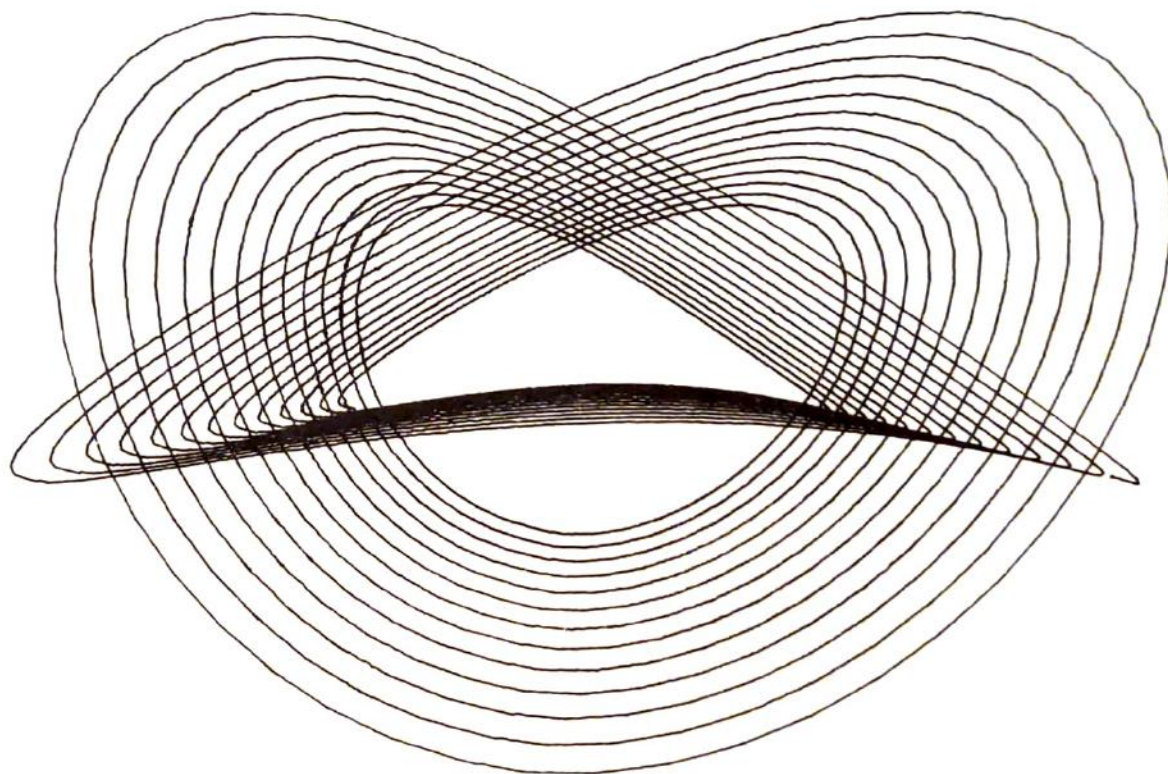
$$w_x = e^{\begin{pmatrix} 0 & I \\ K & 0 \end{pmatrix} t} w_x(0). \quad \text{Similarly,} \quad w_y = e^{\begin{pmatrix} 0 & I \\ K & 0 \end{pmatrix} t} w_y(0).$$

(K is the same for both dimensions since the masses and the lengths are the same.)



In the computation of this function, however, we have completely ignored the effects of friction and air resistance. That is, after one second, the system will have only, say, 99% of the energy it had before, 1% being lost as heat. If the factor of energy remaining after one second is called r , this may be crudely incorporated into the function as:

$$(w_x, w_y) = r^t e^{\begin{pmatrix} 0 & I \\ K & 0 \end{pmatrix} t} (w_x(0), w_y(0))$$
 Computer plots of this function are shown below.



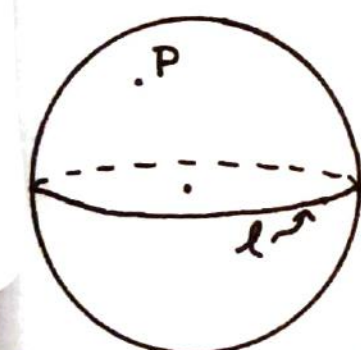
We stated earlier that the simplest type of harmonogram is a Lissajous figure. This is because a Lissajous figure will result if the weight of the lower bob is made close to 0 and the upper and lower bobs are set in straight line motion at right angles to each other. It is because of the number of restrictions imposed by this that the Lissajous figures are relatively uninteresting. The complex beauty that is typical of harmonograms is exemplified by the plot on the cover of this magazine.

NON-EUCLIDEAN GEOMETRY

RICHARD TELLO

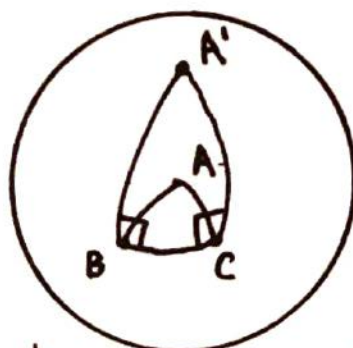
Ever since Euclid first postulated his famed fifth, mathematicians have tried uselessly to prove it as a theorem. If you have forgotten, the fifth goes as follows: Given a line l and a point P not on l there exists one and only one line through P parallel to l . Simple? Not really. Though obvious, many thought it could be proven. Yet many like Saccheri, Lambert, Gauss, Boyalia, and Lobachefsky tried and failed. That was when the idea became not to prove it, but to assume a different possibility and then prove it resulted in a contradiction. Therefore we have three possibilities:

- 1) Through P there exist no lines parallel to l -- the Elliptic hypothesis.
- 2) Through P there exists one and only one line parallel to l -- the Parabolic hypothesis.
- 3) Through P there exist at least two lines parallel to l -- the Hyperbolic hypothesis.

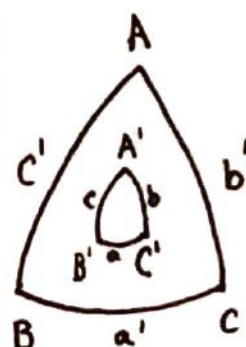


↑ Figure 1 ↑

↓ Figure 2 ↓

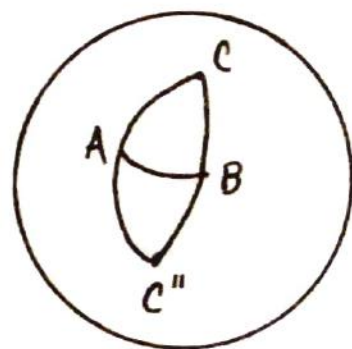


$$\angle A'BC = \angle A'CB = 90^\circ$$



↑ Figure 3 ↑

↓ Figure 4 ↓



ELLIPTIC HYPOTHESIS

A perfect model for this system would be a unit sphere (see fig. 1) where a line is the great circle arc through P .

Now we can start working in this geometry. Note: Any result obtained using our new fifth postulate is invalid in any of the other geometries.

If we are given ABC then let A' , B' , C' (see fig. 2) be one of the poles of each of the sides BC , AC , and AB respectively. $\triangle A'B'C'$ is called a polar triangle of $\triangle ABC$.

Let the polar triangle $A'B'C'$ of ABC be so determined such that it is contained within $\triangle ABC$ (see fig. 3)

Then the two fundamental relations are ;

$$\angle B'AC' + \angle BAC = 180 \quad (1)$$

$$\frac{\angle B'AC'}{180} = \frac{B'C'}{\pi} \quad (2)$$

Dividing (1) by 180 , we have $\frac{\angle B'AC'}{180} + \frac{\angle BAC}{180} = 1$

By substituting (1) into (2) we have ,

$$\frac{\angle BAC}{180} + \frac{B'C'}{\pi} = 1, \text{ or by fig. 3 } \frac{\angle A}{180} + \frac{a}{\pi} = 1 \quad (3)$$

Theorem ; The sum of the lengths of the sides of any triangle in elliptic geometry is less than 2π

Proof : Let C'' be the opposite pole of C (fig. 4).

Then by the triangle inequality for $\triangle ABC'$

$$AB < C''A + C''B$$

$$AB + BC + AC < C''A + AC + C''B + BC = CC'' + CC'' = 2\pi$$

Theorem ; The sum of the angles of a triangle is greater than 180.

Proof : By (3) and symmetry

$$\frac{A}{180} + \frac{a}{\pi} + \frac{B}{180} + \frac{b}{\pi} + \frac{C}{180} + \frac{c}{\pi} = 3,$$

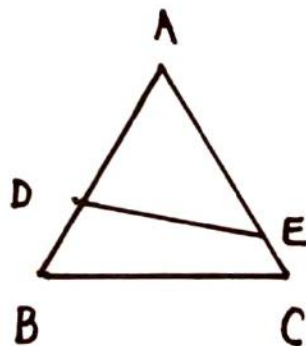
$$\text{or, } \frac{A + B + C}{180} + \frac{a + b + c}{\pi} = 3.$$

Since $a + b + c < 2\pi$, $\angle A + \angle B + \angle C > 180$.

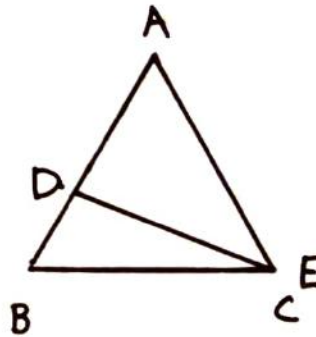
↓ Figure 5 ↓



case 1 ↓



case 2 ↓



case 3 ↓

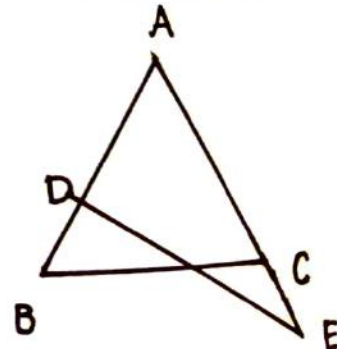


Figure 6 ↑

HYPERBOLIC HYPOTHESIS

As was noted above any result using the Elliptic version or Parabolic version of the fifth is invalid in the Hyperbolic plane (A model would be a pseudosphere or surface of constant negative curvature, see 5). 34

So in the Hyperbolic plane the sum of the angles can't be 180 degrees (Parabolic Hypothesis) and the sum can't be greater than that (Elliptic Hypothesis), so the sum must be less than 180 degrees. From this it is clear that the sum of the angles of a quadrilateral in the hyperbolic plane is less than 4 right angles, which we can now use to prove a new congruence theorem in the Hyperbolic plane.

Theorem: Any AAA correspondence between two triangles in the Hyperbolic plane is a congruence.

Proof: Given $\triangle ABC$ and $\triangle A'B'C'$, let angle $A = \text{angle } A'$, angle $B = \text{angle } B'$, and angle $C = \text{angle } C'$. If we show that no pair of corresponding sides, such as AB and $A'B'$ can be unequal, then it will follow that the two triangles are congruent.

Suppose AB and $A'B'$ are unequal. Then one of them, say AB , must be larger than the other. On AB , measure off AD equal to $A'B'$, and on AC (extend if necessary) measure off AE equal to $A'C'$. Draw DE . Then triangle ADE is congruent to triangle $A'B'C'$ by SAS. Therefore angle $ADE = \text{angle } A'C'B' = \text{angle } ACB$. The rest of the proof hinges on a comparison of the lengths of AE and AC .

Either 1) $AE < AC$ 2) $AE = AC$ or 3) $AE > AC$. The three

possibilities are shown.

Case I: $\angle ADE + \angle BDE + \angle AED + \angle CED = 4$ right angles. Substituting $\angle ABC$ for $\angle ADE$ and $\angle ACB$ for $\angle AED$, we have $\angle ABC + \angle BDE + \angle ACB + \angle CED = 4$ right angles. But this is impossible, because the sum of the angles of a quadrilateral is less than 4 right angles (note if 2 angles are a linear pair then their sum is 2 right angles, this is true in all systems).

Case II: E and C coincide, then $\angle ADE$ is an exterior angle of triangle BDC , and is therefore greater than angle ABC . But we have already established that $\angle ADE = \angle ABC$, so we have a contradiction.

Case III: DE does not cut segment AB . Then by the Pasch axiom, DE cuts BC in some point F . (Pasch axiom: If a line passes through one side of a triangle and through none of the vertices, then it must pass through another side.) Then $\angle ADE$ is an exterior angle of triangle BDF and is greater than angle ABC . Once again we have a contradiction.

Since all possibilities led to contradictions the original assumption was false and so all similar triangles are congruent.

For further reading I suggest; A New Look At Geometry by Irving Adler, College Geometry by David C. Kay, and Non-Euclidean Geometry by Roberto Bologna.

BINOMIAL THEOREM

MICHAEL TRENK

Some people find pleasure in analyzing the first 1000 decimal places of π , e , or 2. Others will search day and night for odd perfect numbers or miracle proofs of Goldbach's Conjecture. I believe in simply enjoying the basic pleasures of mathematics. Understanding and appreciating the full beauty of the Binomial Theorem qualifies as one of the best.

In mathematics, the Binomial Theorem is a way of expanding $(a + b)^n$, an incredible mess but an interesting one nonetheless.

For example: $(a + b)^4 = (a + b)(a+b)(a+b)(a+b)$
 $= aaaa + aaab + aaba + aabb + abaa + abab + abba + abbb$
 $+ baaa + baab + baba + babb + bbaa + bbab + bbba + bbbb$

Note how the Distributive law provides us with a total of $16(2^4)$ terms. Now arrange these 16 terms according to their powers of a and b :

$$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

If we try $(a+b)^n$ (where n is an integer ≥ 0), the result will be:

$$\sum_{i=0}^n \binom{n}{i} a^{n-i} b^i \quad \text{where} \quad \binom{n}{i} = \frac{n!}{i!(n-i)!} \quad \text{if } 0 \leq i \leq n.$$

(Verify this in any good Algebra book)

But do these Binomial coefficients have anything to do with anything else? The answer is an emphatic YES!

Suppose you get married, move to the suburbs, and plan to have 4 children. What would be your chances of ending up with all boys? 3 boys? 2 boys? 1 boy? No boys? (My apologies to feminists) Instead of $(a+b)^4$, try $(b+g)^4$, where B is the probability of a given baby being a boy ($1/2$), and G is the probability of a girl ($1/2$). Since $(1/2 + 1/2)^4 = 1^4 = 1$, this binomial expansion will provide us with all the possible combinations and their probabilities.

$$\begin{aligned} (B+G)^4 &= B^4 + 4B^3G + 6B^2G^2 + 4BG^3 + G^4 \\ &= \frac{1}{16} + 4\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^3 + 6\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^2 + 4\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^3 + \frac{1}{16} \\ &= \frac{1}{16} + \frac{4}{16} + \frac{6}{16} + \frac{4}{16} + \frac{1}{16} \\ &\quad \begin{matrix} 4 \text{ boys,} & 3 \text{ boys,} & 2 \text{ boys,} & 1 \text{ boy,} & 0 \text{ boys} \end{matrix} \end{aligned}$$

Contrary to what many people believe, there is only a $3/8$ chance of getting 2 of each sex. $5/8$ of the time, there will be some imbalance. But don't try this experiment too many times, it causes an exponential population increase! If you were flipping coins or doing anything else with a $1/2$ probability, the same rules would apply.

However, there are some things in life whose probabilities are not $1/2$. Suppose the Yankees (who have a $2/3$ probability of

winning a game) play a four game series. What are their chances of winning 0,1,2,3, or 4 games? This time, we use $(W+L)^4$, where $W = 2/3$ and $L = 1/3$.

$$(W+L)^4 = (2/3)^4 + 4(2/3)^3(1/3) + 6(2/3)^2(1/3)^2 + (2/3)(1/3)^3 + (1/3)^4$$

$$= 16/81 + 32/81 + 24/81 + 8/81 + 1/81$$

$$(4 \text{ wins}) \quad (3 \text{ wins}) \quad (2 \text{ wins}) \quad (1 \text{ win}) \quad (0 \text{ wins})$$

Note that all combinations are possible. The Yankees' $2/3$ probability does not exempt them from losing all 4 games. In fact, they should expect to lose 4 games in a row about twice in a 162 game season. To check the above expansion, get the sum of each probability multiplied by the appropriate number of wins. The result should be $4(2/3)$ or $8/3$.

Casey Stengel, notorious manager of the Mets once remarked, "you can't win 'em all." He knew. But you gotta try pretty hard to lose 'em all too. The Mets, no matter how bad they were, always managed to win at least 40 games. In a binomial distribution, anything can happen, but usually the result will be in the middle, where the probabilities are highest.

Probability is only one application of the Binomial Theorem. What else can it do? Well, it could tell you how many possible ways to walk home through city streets, or how unusual a 7 foot tall person is, or how many distinct rearrangements of the letters in MISSISSIPPI exist!

Here is a facsimile of Pascal's Triangle, a marvellous concoction that makes binomial coefficient calculation easy... and fun too! The i th number in the n th level from the top indicates $\binom{n}{i}$. (Count the first level and number as zero.)

Here are some curious facts about Pascal's Triangle:

Curious Fact #1: $\binom{n}{i} + \binom{n}{i+1} = \binom{n+1}{i+1}$

(father) (mother) (child)

```

      1
     1 1
    1 2 1
   1 3 3 1
  1 4 6 4 1
 1 5 10 10 5 1
etc.

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A new number is born in Pascal's Triangle as the sum of its father and mother immediately above it. The Almighty One Himself sits at the top; below him are two more ones representing Adam & Eve from which the whole species has sprung forth. Since each parent contributes twice to the next generation, we have:

Curious Fact #2: $\sum_{i=0}^n \binom{n}{i} = 2^n$

And, because of the bilateral symmetry in man and many other living creatures:

Curious Fact #3: $\binom{n}{i} = \binom{n}{n-i}$ (Prove this using the definition of $\binom{n}{i}$.)

Curious Fact #4: $\sum_{i=0}^{n/2} \binom{n}{2i} = \sum_{i=0}^{(n/2)-1} \binom{n}{2i+1}$ (prove by tracing the ancestry of each set of terms.)

But what is the destiny of the sprawling mass of coefficients produced by $(a+b)^n$ as n increases? First consider the plight of $\binom{2n}{n}$, the middle (and greatest) term. 2 generations later, the middle term $\binom{2n+2}{n+1}$ will be exactly $\frac{4n+2}{n+1}$ times as large as $\binom{2n}{n}$.

Since total population has quadrupled during that time, the middle term now represents a smaller percentage of the population. This ratio, $\binom{2n}{n} / 2^{2n}$ is also equal to $\frac{1 \cdot 3 \cdot 5 \cdot 7 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdot 8 \cdots 2n}$, which

approximates $1/\sqrt{2\pi n}$. Sadly, there is no positive limit. The ratio will just sink lower and lower, never hitting zero, but never satisfied with any positive value.

What about the other friendly numbers? If the middle term is being dwarfed by 2^{2n} , the others don't even have a chance alone. High valued elements occur further and further from the center, but this spread is overshadowed by the faster spread of relatively small numbers on the outside. The result is that a profile on absolute scale (see graph) will appear to be more and more squashed.

This is the concept of "The Law of Averages", sometimes informally known as "in the long run, everything will work out", "life has its ups and downs", etc. What it all means is basically the more trials or experiments attempted, the more likely an observed probability will be closer to "normal".

Looking at the Binomial Distribution Curves from a logarithmic point of view provides a very different perspective. $y = \log n x$ approximates a perfect parabola which grows taller at a slightly less than linear rate as n increases. The curves are not subject to squeezing.

Finally, I must show you the infinite Binomial Distribution Curve, known as the "Bell Curve". It is really the same shape as the finite curves; I have unsqueezed it, and adjusted its proportions to a "standard" size. The center is at $x=0$, the area over the x axis is exactly 1, and its points of inflection are strategically positioned at $x=\pm 1$. The equation is:

$$y = e^{-x^2} / \sqrt{\pi}$$

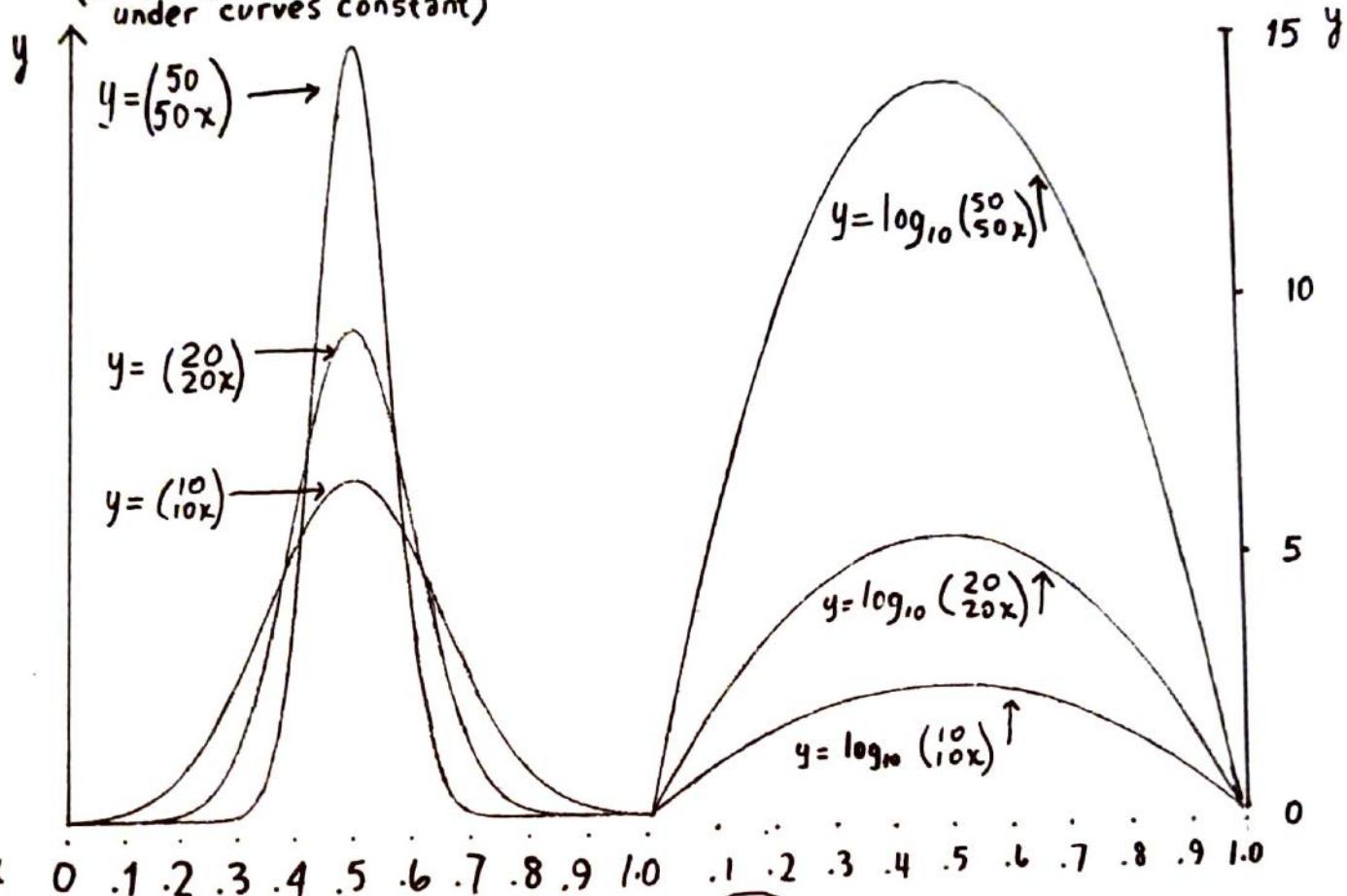
I hope you have enjoyed this introduction to the world of binomials; there are plenty of books in the Stuyvesant library on this (and many other) topics, eager to be read. And remember- the better part of your life depends on the binomial theorem. Knowing and loving the binomial theorem is knowing and loving yourself.

The Binomial Distribution:

Absolute Scale

(scale adjusted to make areas under curves constant)

Logarithmic Scale



Standard "Bell" Curve

$$y = \frac{e^{-x^2}}{\sqrt{\pi}}$$

area = 1

← point of inflection
 $x = 1$

-2.5 -2 -1.5 -1 -0.5 0 .5 1 1.5 2 2.5

NEW YORK UNIVERSITY
DEPARTMENT OF MATHEMATICS

PRESENTS:

.....SOME BRAIN TEASERS TO TEST YOUR METTLE. A SPLENDID AWARD
WILL BE MADE FOR THE BEST EARLIEST SOLUTIONS. GOOD LUCK!!

1. Let $f(x)$ be a cubic polynomial with real coefficients such that the real numbers $x_1 < x_2 < x_3 < x_4 < x_5$ satisfy

(a) $f(x_1) + f(x_2) + f(x_3) > 0$

(b) $f(x_2) + f(x_3) + f(x_4) < 0$

(c) $f(x_3) + f(x_4) + f(x_5) > 0$

Show that f has three real zeroes.

2. A sequence $\{u_n\}$, $n = 0, 1, 2, \dots$ satisfies $u_0, u_1 > 0$,
 $u_n = 3u_{n-1} + 5u_{n-2}$ for $n \geq 2$. Find the smallest integer
 M such that for all sufficiently large n

$$|u_n| < M^n.$$

3. Applying the Σ operator to a sequence $a_1, a_2, a_3, \dots$ means forming the new sequence $a_1, a_1+a_2, a_1+a_2+a_3, \dots$. Let k be any positive integer ≥ 2 . Take the sequence of positive integers. Delete every k -th element, apply the summation operator, delete every k -1st element, apply the summation operator again etc., until the summation operator has been applied $k-1$ times. Prove that the final sequence is $1, 2^k, 3^k, 4^k, \dots$

Send your solutions to: DEPARTMENT OF MATHEMATICS
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DEADLINE:

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